

自然语言处理

WELCOME!

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2025-2026-1 学期

基于 Julia, Neubig 等课件更新

面试问题

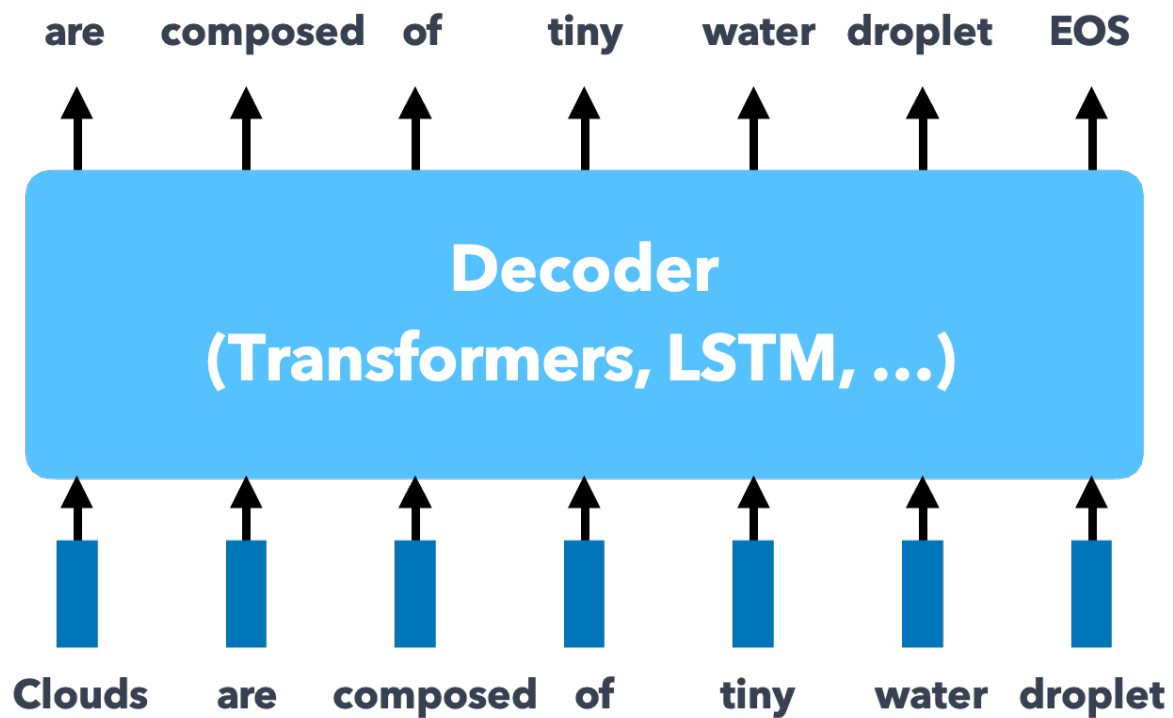
- 什么是提示词工程，跟In-context Learning的关系和区别
- 对比**零样本提示**、**少样本提示**、**微调**三种方式在“特定领域任务（如医疗病历分析）”中的优缺点，并说明在什么情况下优先选择提示词工程而非微调？
- 在大模型生成长文本（如小说章节、调研报告）时，经常出现“上下文不一致、逻辑断裂”的问题。请结合提示词工程的技巧，提出至少 3 种优化方案，并举例说明。

大纲

- 提示词与In-Context Learning
- 更具策略性的提示词方案
- 高级提示词技术

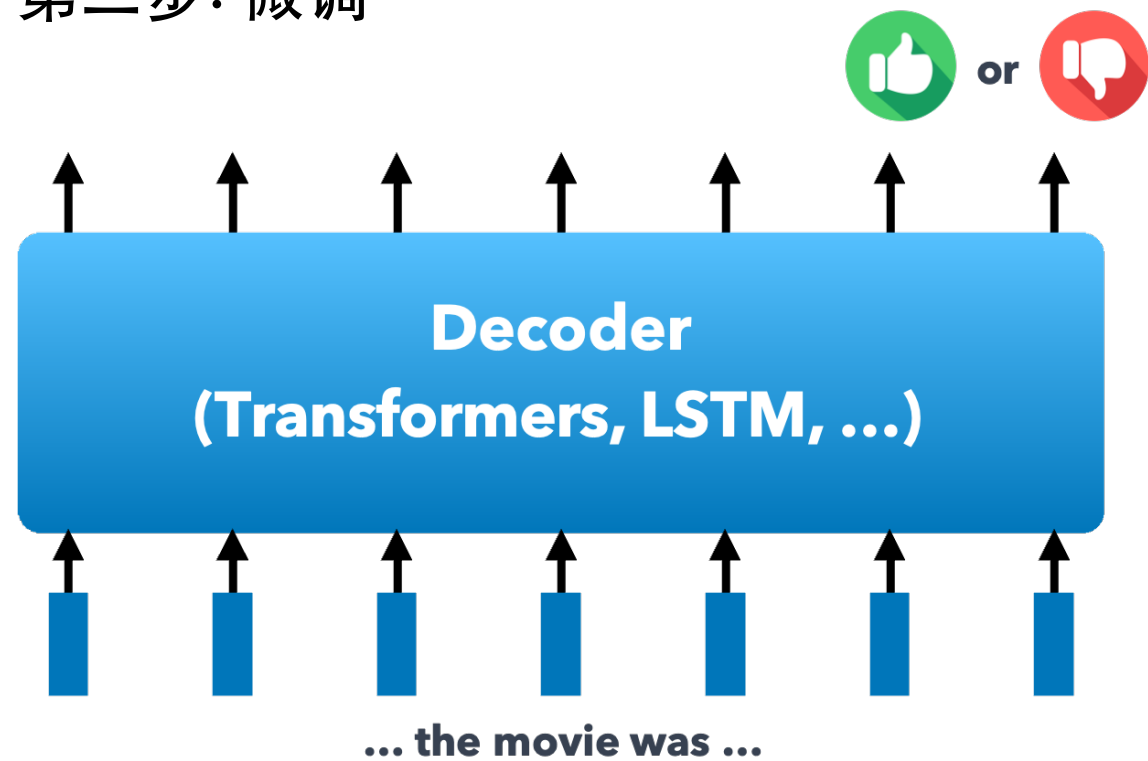
针对特定任务的监督微调 Supervised Fine-tuning

- 第一步: 预训练



海量数据: 学习通用语言

- 第二步: 微调

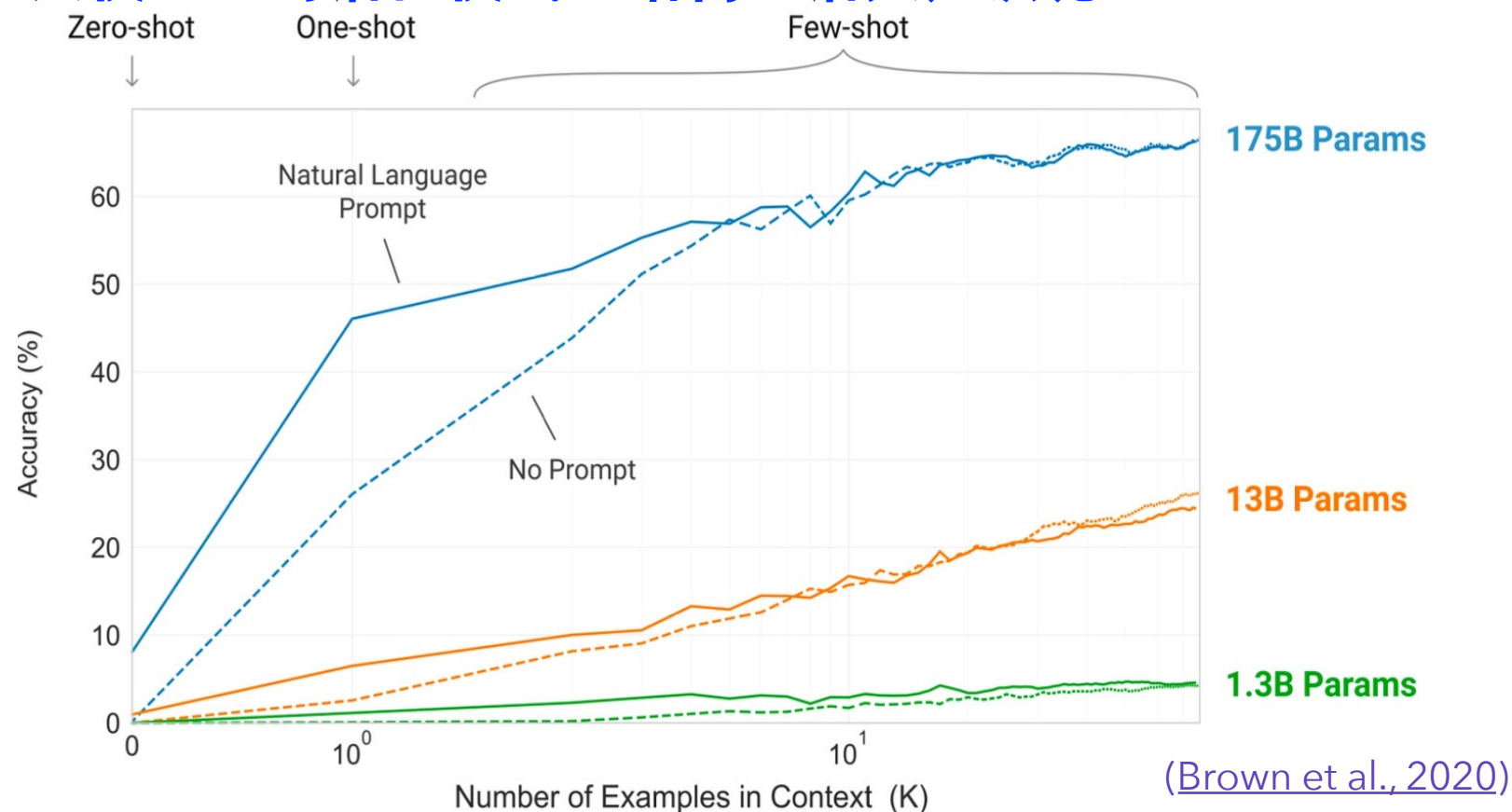


少量数据: 适配任务

我们可以不经微调完成任务吗？

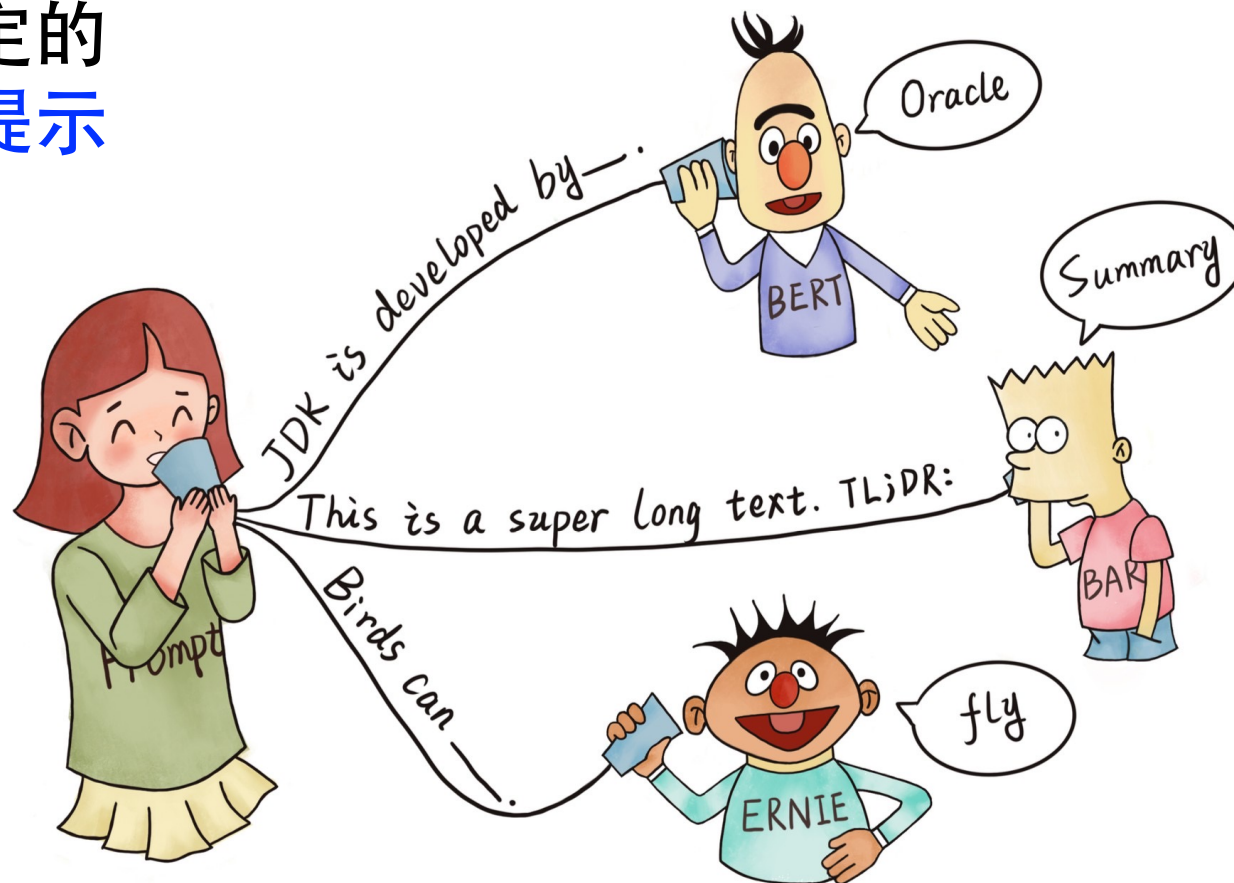
- 如何利用已蕴藏在大模型里的语言模式、结构、语义知识呢？

当进入
GPT3...



什么是提示词? Prompting

- 鼓励预训练大模型解决一定的任务通过写特定的文本“**提示词**”



基础提示词：情感分析

- 将文本(y)附于句子后，引导/elicit模型完成任务



Food Review: "I recently had the pleasure of dining at Fusion Bites, and the experience was nothing short of spectacular. The menu boasts an exciting blend of global flavors, and each dish is a masterpiece in its own right."



提示词引导情感分析: = Overall, I think the restaurant is _____.

模型完成



awesome

fantastic

great

good

GPT-3

[[Brown et al., 2020](#)]

- **方案**: 如果我们将自回归模型扩大10倍，会怎么样？
- **结果**: 语言模型可以做In-Context Learning (ICL)

Language Models are Few-Shot Learners

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on

In-Context Learning (ICL)/上下文学习:
“在**无监督预训练/unsupervised pre-training**阶段，语言模型会习得**相关能力**；并在**推理阶段**利用这些能力，**快速适配**或识别目标任务。我们使用‘上下文学习’这一术语，来描述该过程的内层循环——这一循环在每一条序列的前向传播过程中完成。”

we distinguish from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

[Brown et al., 2020]

Learning via SGD during unsupervised pre-training

inner loop

Addition

1	5 + 8 = 13
2	7 + 2 = 9
3	1 + 0 = 1
4	3 + 4 = 7
5	5 + 9 = 14
6	9 + 8 = 17

↑
sequence #1

In-context learning

Rearrange Words

1	gaot => goat
2	sakne => snake
3	brid => bird
4	fsih => fish
5	dcuk => duck
6	cmihp => chimp

↑
sequence #2

In-context learning

French Translate

1	thanks => merci
2	hello => bonjour
3	mint => menthe
4	wall => mur
5	otter => loutre
6	bread => pain

↑
sequence #3

In-context learning

In-Context Learning (ICL)

[[Brown et al., 2020](#)]

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French:
2 cheese => .....
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

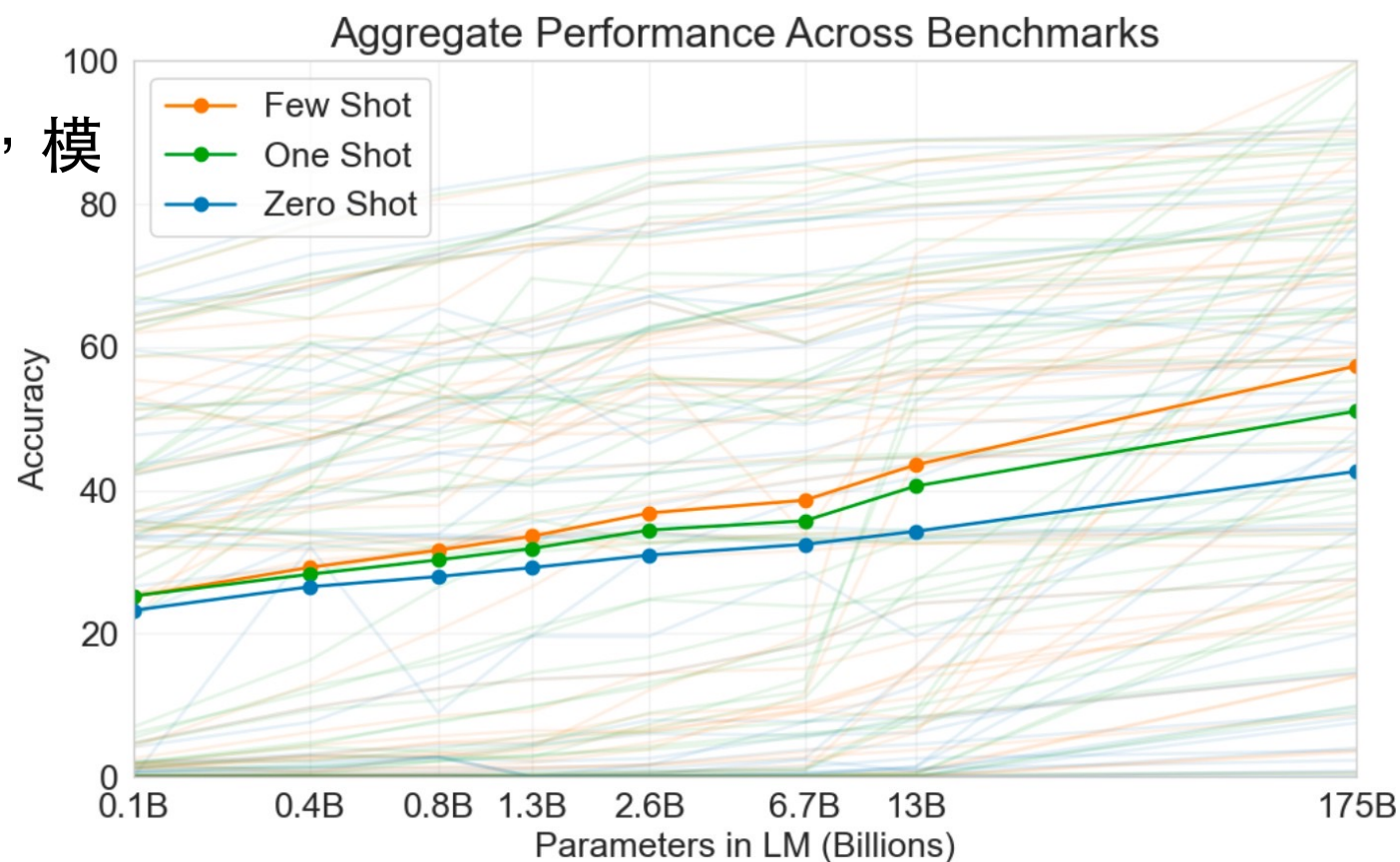
```
1 Translate English to French:
2 sea otter => loutre de mer
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => .....
```


零样本学习 vs. 少样本学习

Zero-shot vs. Few-shot

[[Brown et al., 2020](#)]

- 一般而言，少样本示例的效果要优于零样本设置
- 但这并不意味着示例数量越多，模型性能就一定会越好



ICL的困难

- 对上下文示例敏感
 - 次序的例子[[Lu et al. 2021](#)]

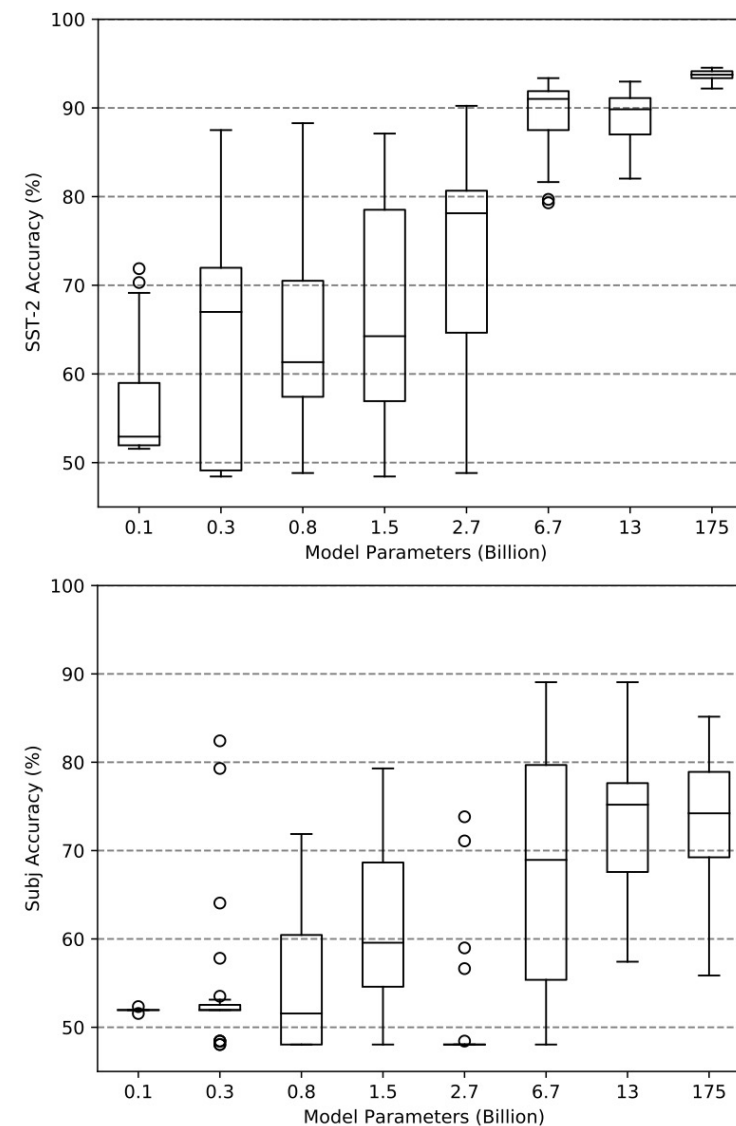


Figure 1: Four-shot performance for 24 different sample orders across different sizes of GPT-family models (GPT-2 and GPT-3) for the SST-2 and Subj datasets.

ICL的困难

- 对上下文示例敏感
 - 标签平衡和覆盖度 [Zhang et al. 2022]

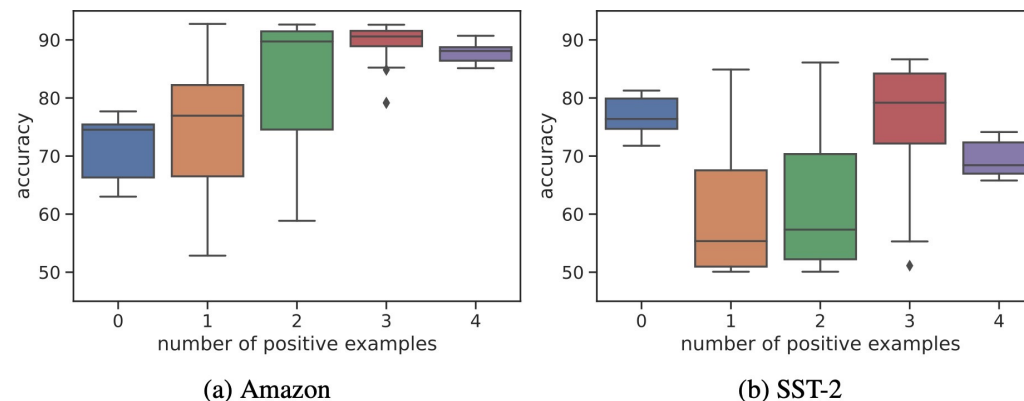


Figure 3: Accuracies of Amazon and SST-2 with varying **label balance** (number of positive examples in demonstration), across 100 total random samples of 4 demonstration examples.

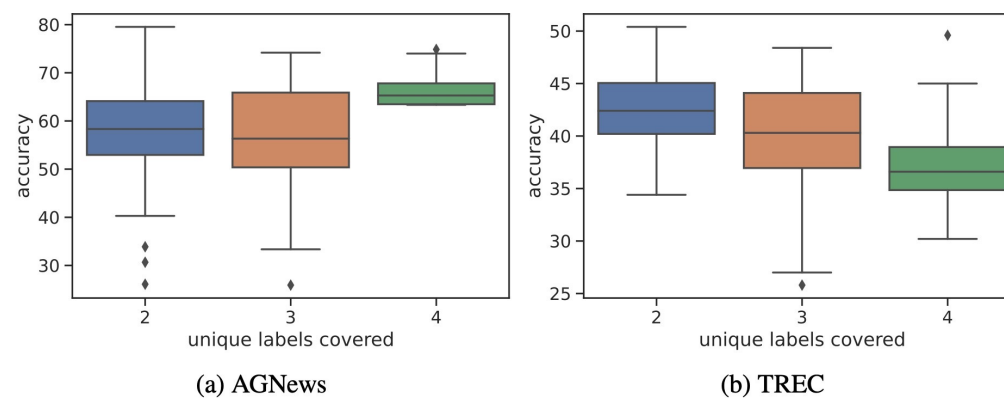


Figure 4: Accuracies of AGNews and TREC with varying **label coverage** (number of unique labels covered in demonstration), across 100 total random samples of 4 demonstration examples. Demonstration set that only covers 1 label is very unlikely and does not appear in our experiments.

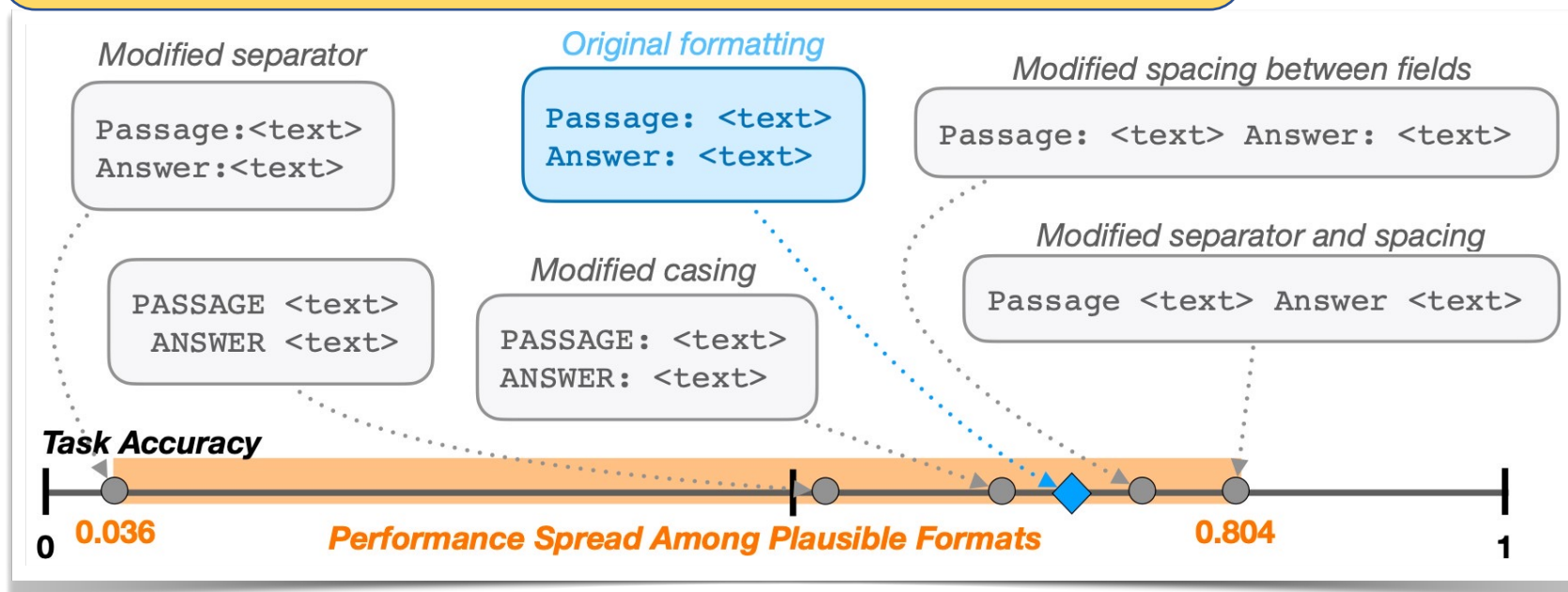
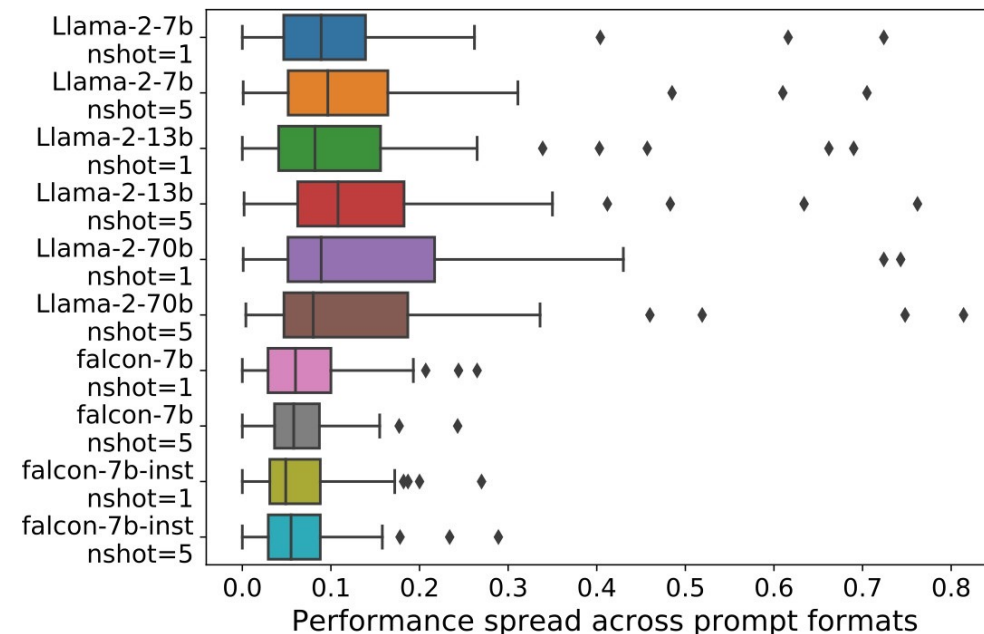
ICL的困难

- 对提示词格式的敏感 [Sclar et al, 2023]

给定合理的格式 $p_1, \dots, p_n$, 数据集 D 以及标量指标 m

性能区间: $[\min_i m(p_i, D), \max_i m(p_i, D)]$

性能波动值 (spread): $\max_i m(p_i, D) - \min_i m(p_i, D)$



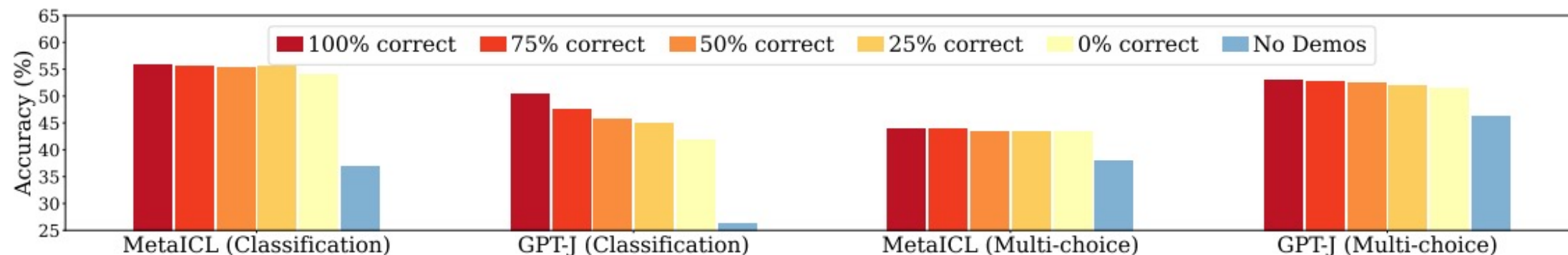
ICL的困难

- 对提示词格式的敏感 Sensitive to prompt forms [[Sclar et al., 2023](#)]
- 对上下文例子的敏感 Sensitive to in-context examples
 - 例子的次序 Example ordering [[Lu et al. 2021](#)]
 - 标签平衡与覆盖度 Label balance & coverage [[Zhang et al. 2022](#)]
- 跨任务的性能不一致
 - 任务性能在一定程度上取决于训练数据中是否存在类似的任务
- 要预先判断一个模型能否稳定完成某项任务并非易事（除非实际测试验证）

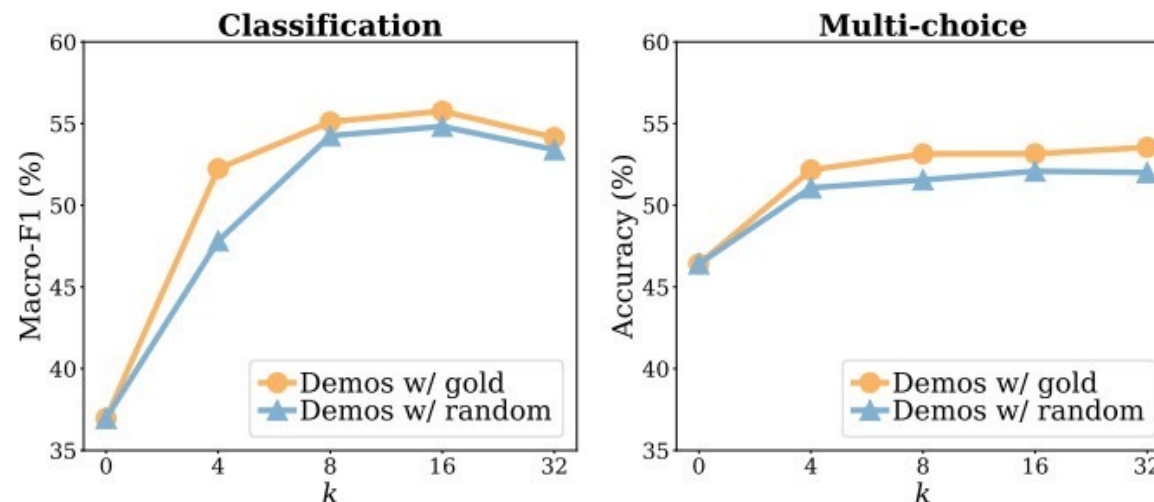
ICL的一些特性

[[Min et al. 2022](#)]

- 用随机标签替换正确的标签有时几乎不会影响性能准确性



- 更多的例子有时会损害模型性能



提示词工程技巧

Tips on Prompt Engineering

- Good Tutorial: <https://www.promptingguide.ai/introduction/tips>
- 实验是关键
 - 最佳提示词通过测试和提炼中进化而来
- 要清晰、具体且简洁 / Be Clear, Specific, and Concise
 - Direct and relevant instructions yield the best results
- 迭代和改进 / Iterate & Improve
 - 每次尝试都提供了对任务提示进行微调的洞见

跟人类类似，但大语言模型在指令不明确时不会抱怨

方法比较

	经典深度学习方法	预训练/微调方法	上下文学习
训练数据量	> 1百万 (理想下)	>1万	0~20
是否需要预训练	否	是	是
是否需要在训练样本上更新梯度	是	是	否
“学习” 从何而来？	数据的统计	从预训练的表征，迁移到训练数据的统计	来自预训练的语言表征、任务描述及示例(形式不限)

更具策略性的提示词方案

思维链推理等

思维链提示词

Chain-of-Thought Prompting

[[Wei et al. 2022](#)]

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

思维链的优势

- 分解复杂任务

- 将多步骤问题拆解为若干中间步骤，从而实现更详尽的逐步推理

- 提升可解释性

- 能够清晰呈现模型的决策路径，有助于定位并修正推理过程中的错误

- 广泛适用性

- 适用于数学问题求解、常识推理、符号操作等各类任务 — 本质上，任何可通过基于语言的推理完成的任务均可使用

- 易于实现

- 在少样本学习中，只需在提示词中加入逐步推理的示例，即可在大语言模型中实现该功能

无监督思维链提示词

[[Kojima et al. 2022](#)]

- 只需添加一个能够引导模型解释其决策过程的提示词，就能触发模型的推理能力

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

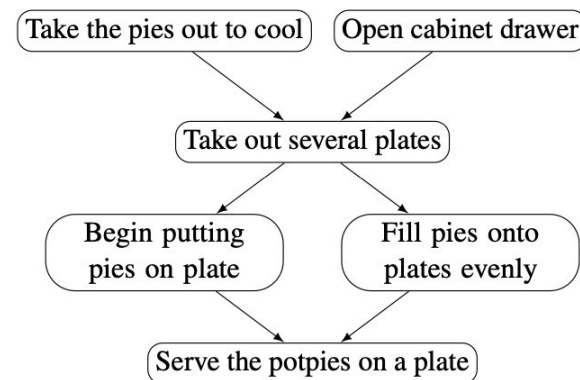
(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

注：如今即便不添加专门指令，GPT 系列模型也能生成推理链（这一能力很可能源于指令微调/instruction tuning）

程序结构化提示法

[[Madaan et al. 2022](#)]

- 在预测结构化输出时，使用编程语言而非自然语言往往能提升准确率
- Why?** 程序具备高度结构化的特性，且这类结构化数据已被纳入预训练语料
- 要求模型生成 JSON 格式的输出，能够有效解决格式规范相关的问题



(a) The script $\mathcal{G}$

```
class Tree:

    goal = "serve the potpies on a plate"

    def __init__(self):
        # nodes
        take_pies_out_to_cool = Node()
        open_cabinet_drawer = Node()
        take_out_several_plates = Node()
        ...
        # edges
        take_pies_out_to_cool.children =
            [take_out_several_plates]
        open_cabinet_drawer.children =
            [take_out_several_plates]
        ...
```

(b) $\mathcal{G}$ converted to Python code $\mathcal{G}_c$ using our approach

```
digraph G {
    begin -> take_pies_out_to_cool;
    begin -> open_cabinet_drawer;
    take_pies_out_to_cool ->
        take_out_several_plates;
    open_cabinet_drawer ->
        take_out_several_plates;
    take_out_several_plates ->
        begin_putting_pies_on_plates;
    begin_putting_pies_on_plates ->
        serve_potpies_on_plate;
    fill_pies_onto_plates_evenly ->
        serve_potpies_on_plate;
    serve_potpies_on_plate -> end;
}
```

(c) Straightforward encodings of the graph using the “DOT”

```
[
    (take_pies_out_to_cool,
     take_out_several_plates),
    (open_cabinet_drawer,
     take_out_several_plates),
    (take_out_several_plates,
     begin_putting_pies_on_plates),
    (take_out_several_plates,
     fill_pies_onto_plates_evenly),
    (begin_putting_pies_on_plates,
     serve_potpies_on_plate),
    (fill_pies_onto_plates_evenly,
     serve_potpies_on_plate),
    (serve_potpies_on_plate, end)
]
```

(d) Text format, or as a list of edges (node pairs)

程序辅助语言模型

- 使用程序生成输出，往往比自然语言输出更为精准
- 这一点在处理数值类问题时尤为实用
- 典型案例可参考 ChatGPT 代码解释器、Bard 代码执行功能
- 该方法在智能体/工具调用的相关应用中，同样具有极高的实用价值

[Gao et al. 2022]

Chain-of-Thought (Wei et al., 2022)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 tennis balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves of bread did they have left?

Model Output

A: The bakers started with 200 loaves. They sold 93 in the morning and 39 in the afternoon. So they sold $93 + 39 = 132$ loaves. The grocery store returned 6 loaves. So they had $200 - 132 - 6 = 62$ loaves left. The answer is 62.



Program-aided Language models (this work)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 tennis balls.
`tennis_balls = 5`
2 cans of 3 tennis balls each is
`bought_balls = 2 * 3`
tennis balls. The answer is
`answer = tennis_balls + bought_balls`

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves of bread did they have left?

Model Output

A: The bakers started with 200 loaves
`loaves_baked = 200`
They sold 93 in the morning and 39 in the afternoon
`loaves_sold_morning = 93`
`loaves_sold_afternoon = 39`
The grocery store returned 6 loaves.
`loaves_returned = 6`
The answer is
`answer = loaves_baked - loaves_sold_morning`
`- loaves_sold_afternoon + loaves_returned`

```
>>> print(answer)
74
```



ReAct 提示词

- 交替进行**推理与行动**
- 通过逐步推理问题来“**出声思考**”
- 基于推理结果执行**特定行动**，例如检索更多信息或核验事实
- 形成类人的**交互式**任务求解路径，具备更强的**可解释性**

Aside from the Apple Remote, what other devices can control the program Apple Remote was originally designed to interact with?

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: **Search**[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the **Front Row media center** program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search **Front Row** next and find what other device can control it.

Act 2: **Search**[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', ' **Front Row (software)** ', ...]

Thought 3: Front Row is not found. I need to search **Front Row (software)** .

Act 3: **Search**[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: **Front Row (software)** is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

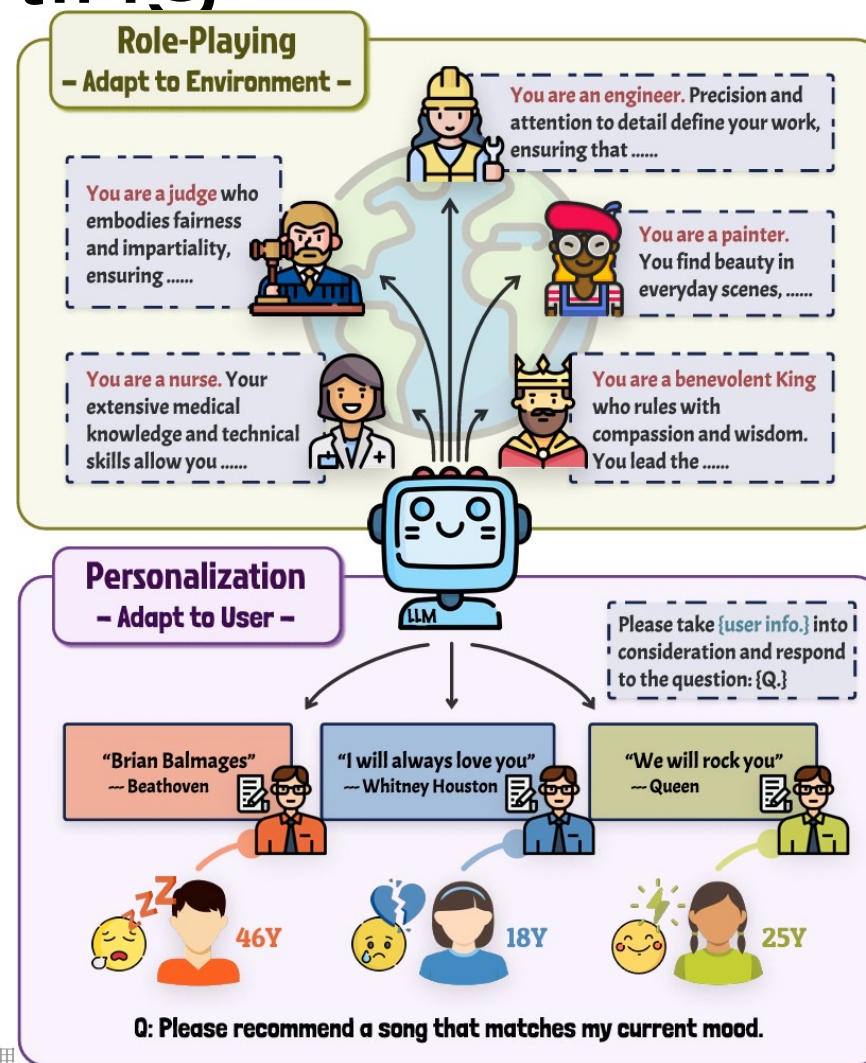
Act 4: **Finish**[**keyboard function keys**]



角色设定提示词 Persona-Based Prompting

[Tseng et al. 2024]

- **角色扮演**: 语言模型在设定的环境中，依照分配的角色身份展开输出
- **个性化定制**: 语言模型结合用户的角色特征，生成贴合需求的针对性回复
- **优势**: 提升交互参与度，输出兼具专业性与场景适配性的回复内容
- **应用场景**: 推荐系统、客户服务，以及医疗、法律等专业领域



自动提示词生成方法

Automatic Prompting Methods

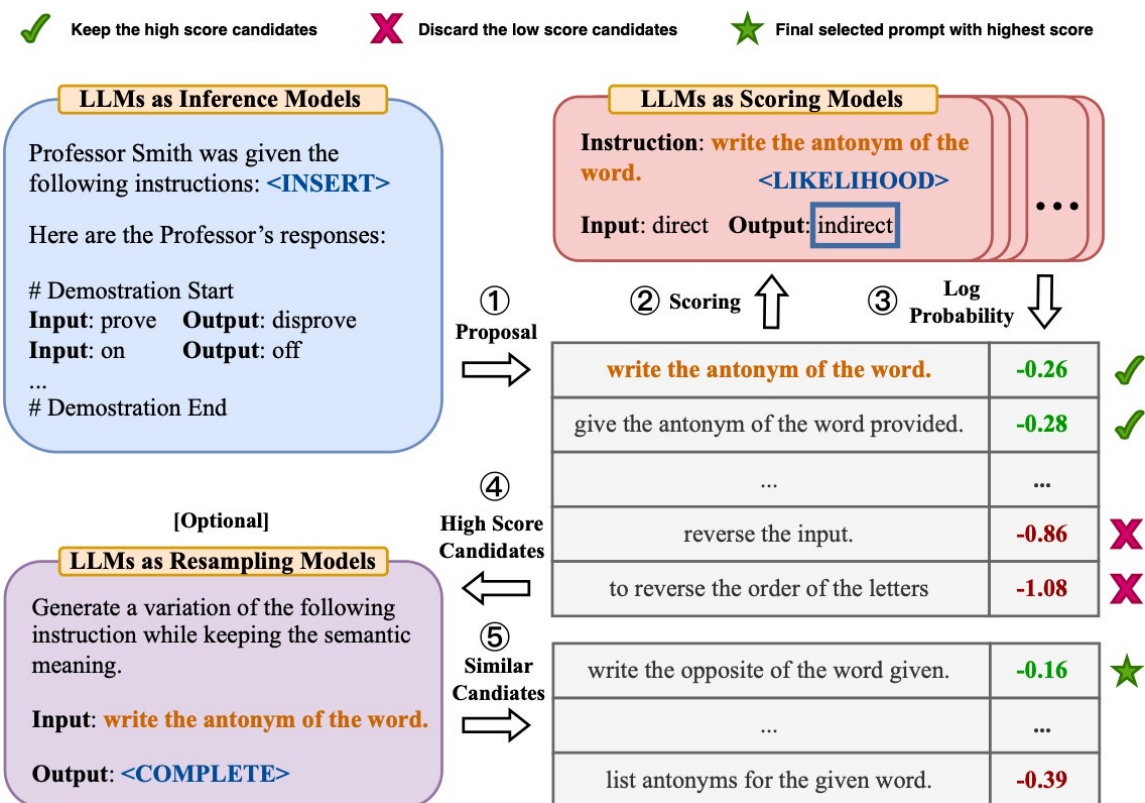
- 自动改写/Automatic Paraphrases
- 迭代优化/Iterative Refinement
- 基于梯度的搜索/Gradient-based Search
- (相关技术)参数高效微调/(Related) Parameter Efficient Fine-tuning
 - 前缀微调/Prefix-tuning, 提示微调/prompt-tuning
- ...

提示词改写法

Prompt Paragraphing



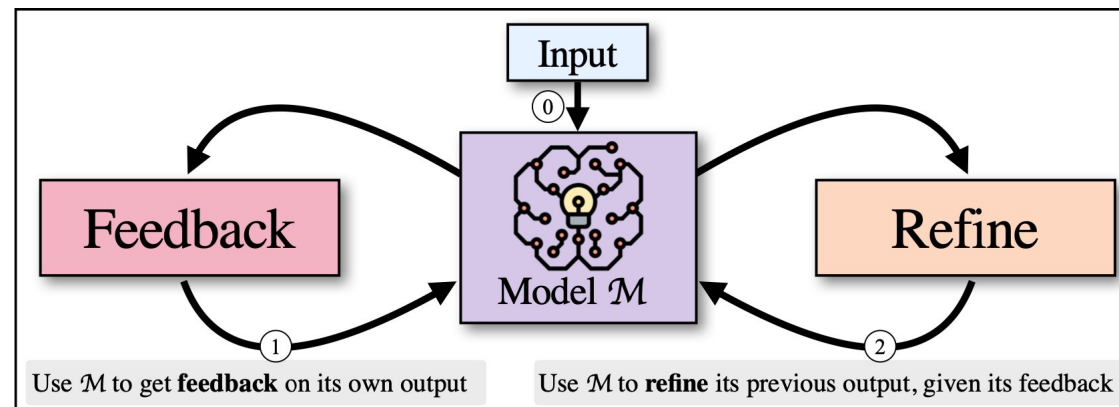
- 对现有提示词进行改写，以生成其他候选版本[Jiang et al, 2019]
- 这一过程可通过迭代改写的方式实现[Zhou et al, 2021]



自优化提示词 Self-Refinement Prompting

[[Madaan et al. 2023](#)]

- **定义**: 当语言模型接收到自身生成的反馈信息时，会对其输出结果进行修正



(a) Dialogue: x, y_t

(b) FEEDBACK fb

(c) REFINE y_{t+1}

- **优势**
 - **实现简便**: 仅需单个模型即可同时完成生成与优化两项任务
 - 降低最终输出中出现错误信息或幻觉内容的概率

User: I am interested in playing Table tennis.
Response: I'm sure it's a great way to socialize, stay active

(d) Code optimization: x, y_t

```
Generate sum of 1, ..., N
def sum(n):
    res = 0
    for i in range(n+1):
        res += i
    return res
```

Engaging: Provides no information about table tennis or how to play it.
User understanding: Lacks understanding of user's needs and state of mind.

(e) FEEDBACK fb

This code is slow as it uses brute force. A better approach is to use the formula ... $(n(n+1))/2$.

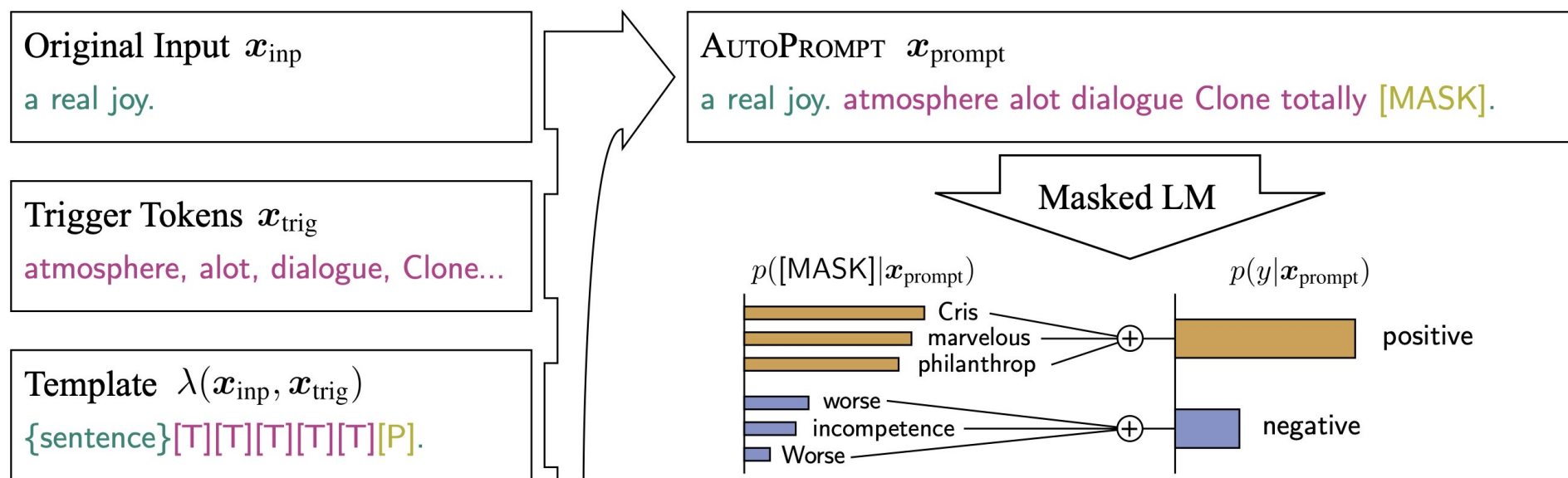
Response (refined): That's great to hear (...) ! It's a fun sport requiring quick reflexes and good hand-eye coordination. Have you played before, or are you looking to learn?

(f) REFINE y_{t+1}

```
Code (refined)
def sum_faster(n):
    return (n*(n+1))/2
```

基于梯度的搜索 Gradient-Based Search

[[Shin et al. 2020](#)]

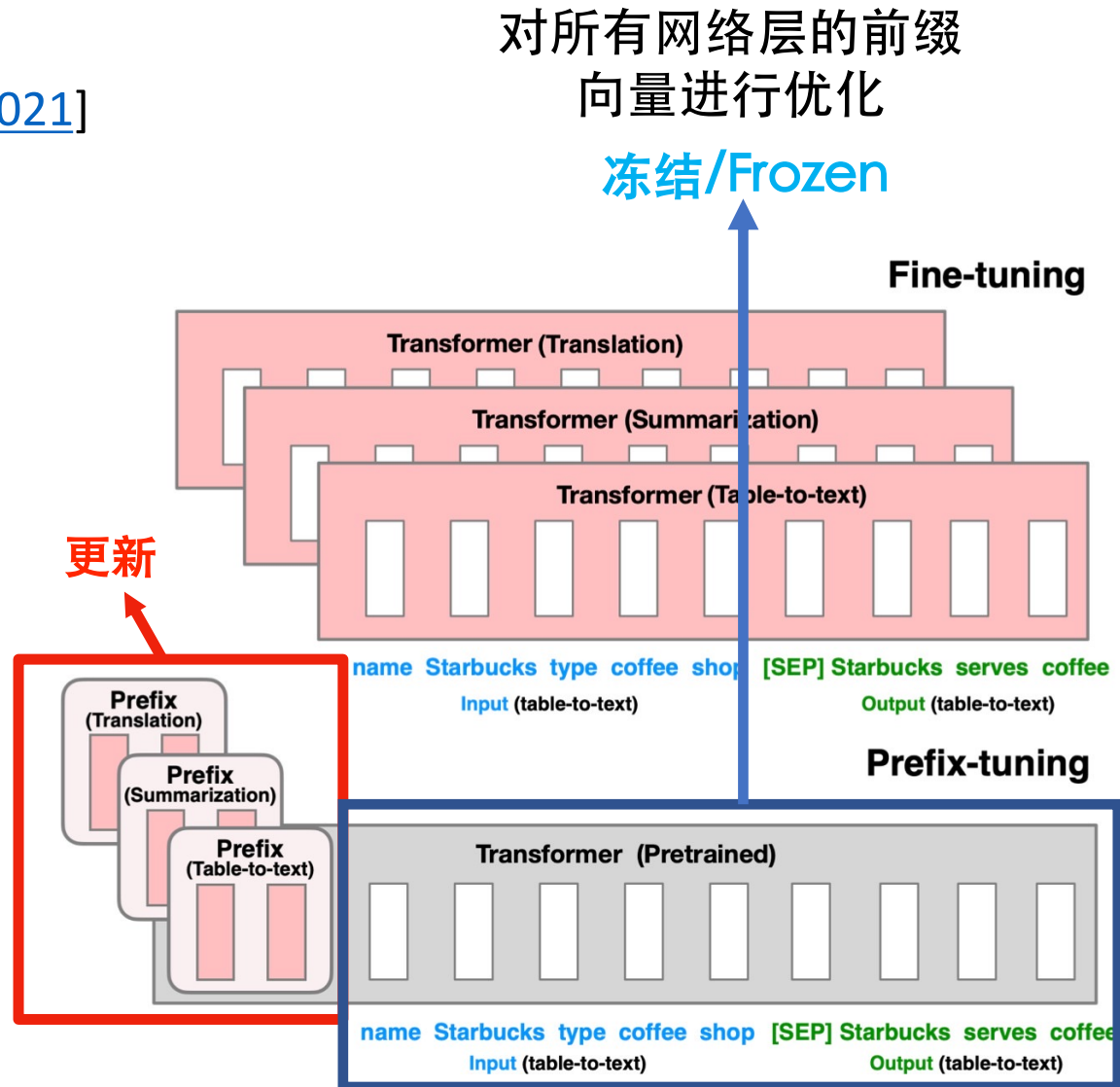


- **优势**: 针对目标输出明确的任务，可实现完全自动化
- **劣势/Downside**:
 - 通常不具备可解释性(优化得到的提示词可能语义混乱)
 - 优化过程耗时较长，对于大型模型甚至可能无法完成优化

前缀微调 Prefix-Tuning

[[Li and Liang, 2021](#)]

- 在保持预训练语言模型参数冻结的前提下，为**每个Transformer块**学习一个小型的、任务专属的连续向量(该向量被称为前缀)
- 仅需微调模型 0.1% 的参数，就能达到与全量微调相当的效果，在低数据场景下表现尤为突出

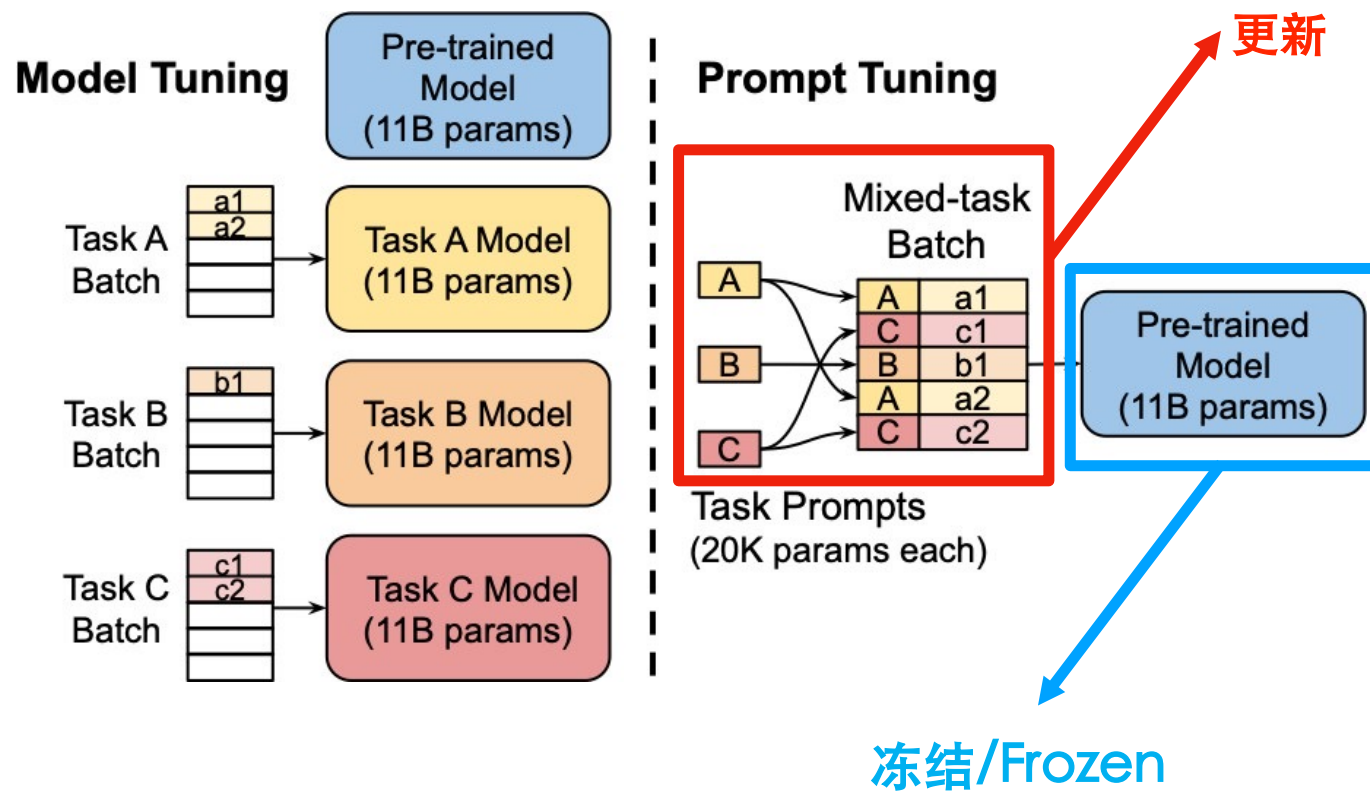


提示词微调 Prompt-Tuning

[Lester et al., 2021]

仅对**嵌入层**进行优化

- 与前缀微调同期提出的研究方案
- 在编码器侧，将单个“软提示/soft prompt”表征前置添加至**嵌入层**输入中
- 所需优化的参数量**少于**前缀微调



提示词工程进阶内容

- 其他主流提示词技术可参见以下链接
 - <https://www.promptingguide.ai/techniques>
- 综述文献: [Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing](#)

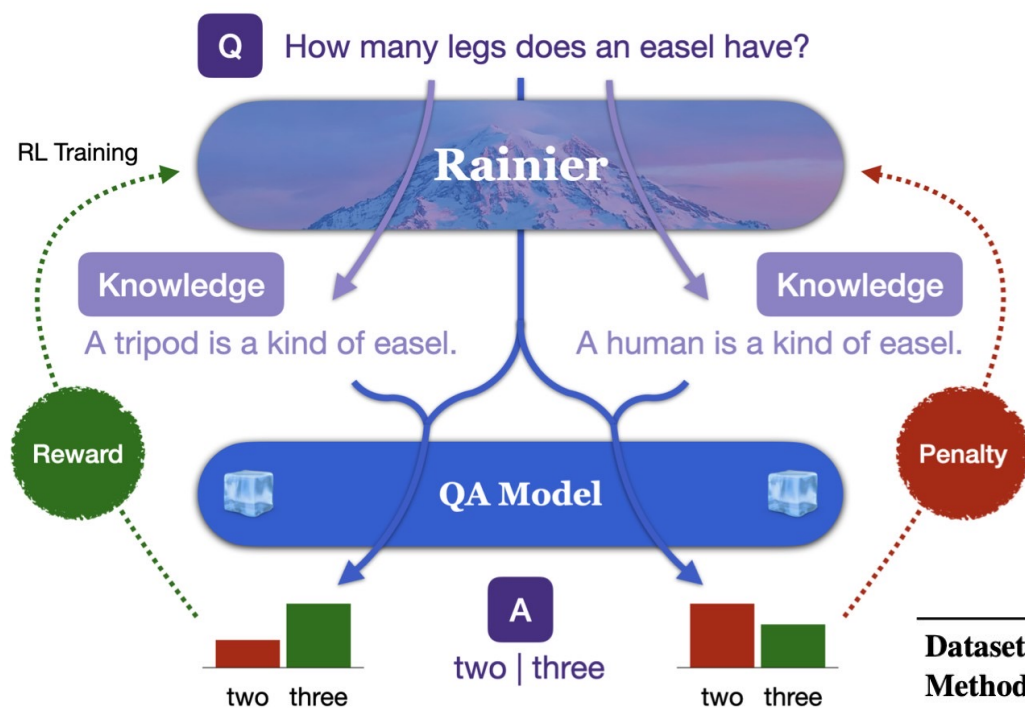
高级提示词技术

1. 知识增强推理与对话式 Knowledge Enhanced Reasoning & Dialog
2. “先思后述” Think-Before-Speaking
3. 智能体与工具调用 Agent & Tool Use
4. 基于澄清式提问的偏好提取 Preference Elicitation with Clarification Questions

知识增强推理

Knowledge Enhanced Reasoning

[Rainier, Liu et al. 2022]

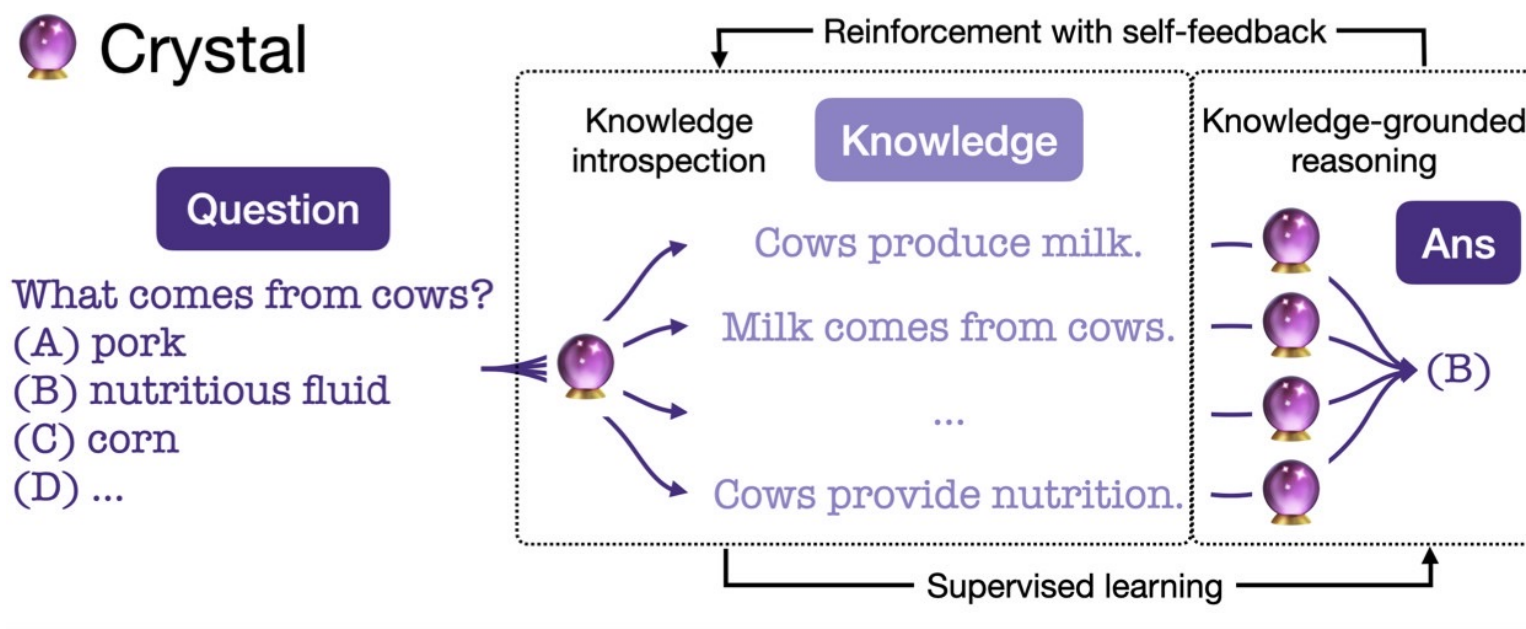


Dataset → Method ↓	CSQA		QASC		PIQA		SIQA		WG		Avg.	
	dev	test	dev	test	dev	test	dev	test	dev	test	dev	test
UQA-large (0.77B)	61.43	53.00	43.09	45.65	63.66	65.50	53.84	57.21	53.35	54.67	55.07	55.21
+ Few-shot GPT-3-Curie (13B)	66.34	–	53.24	–	64.25	–	58.29	–	55.56	–	59.54	–
+ Self-talk GPT-3-Curie (13B)	63.31	–	49.89	–	65.23	–	51.89	–	52.96	–	56.66	–
+ DREAM (11B)	64.54	–	49.46	–	64.74	–	51.59	–	56.12	–	57.29	–
+ RAINIER-large (0.77B) [ours]	67.24	60.18	54.97	54.13	65.67	67.09	57.01	59.01	56.91	57.39	60.36	59.56

知识增强推理

Knowledge Enhanced Reasoning

[[Crystal, Liu et al. 2023](#)]



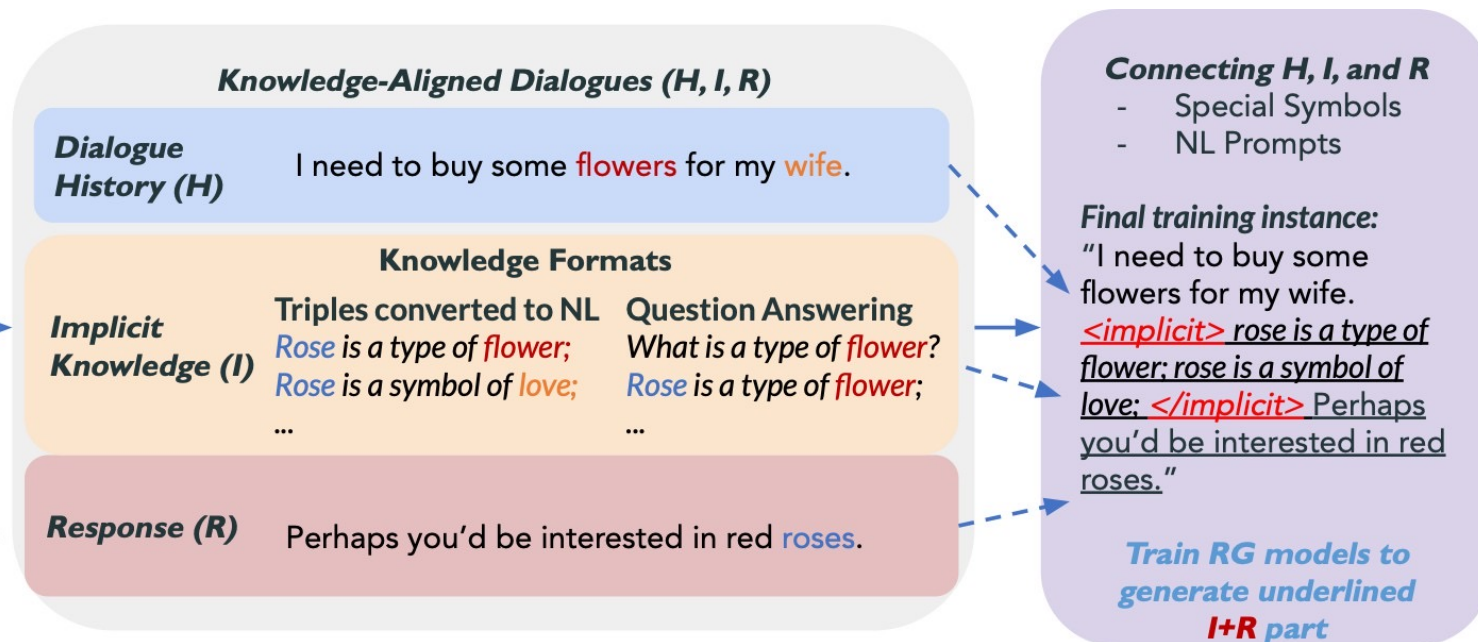
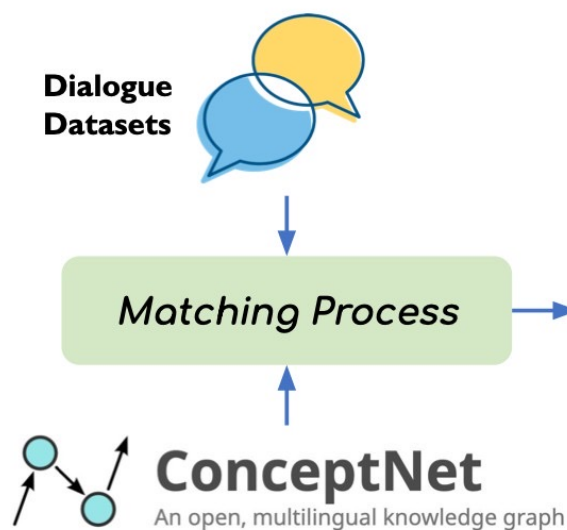
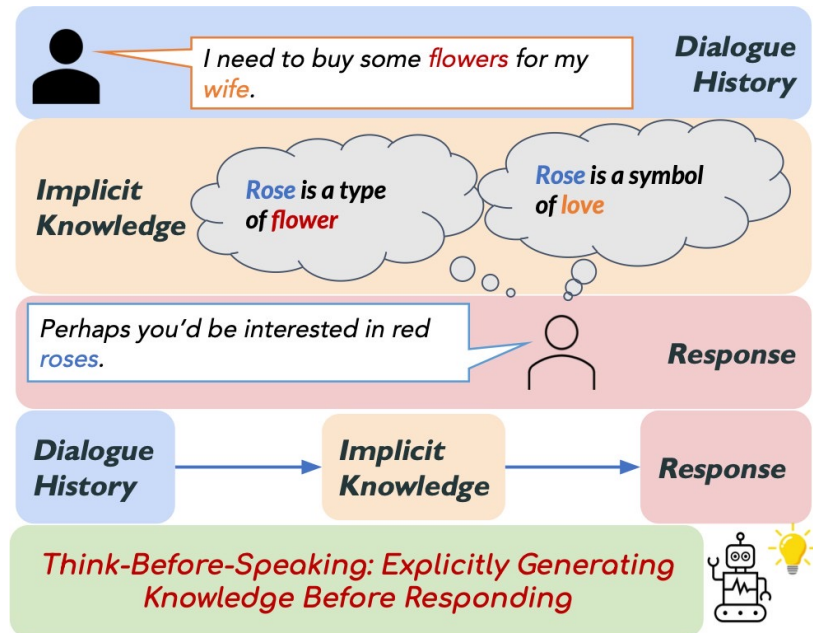
Chain-of-Thought

Question

GPT-3.5

The answer must be something that comes from cows. Of the above choices, only nutritious fluid (B) is something that comes from cows. So the answer is (B).

知识增强对话 Knowledge Enhanced Dialog [Zhou et al. 2023]

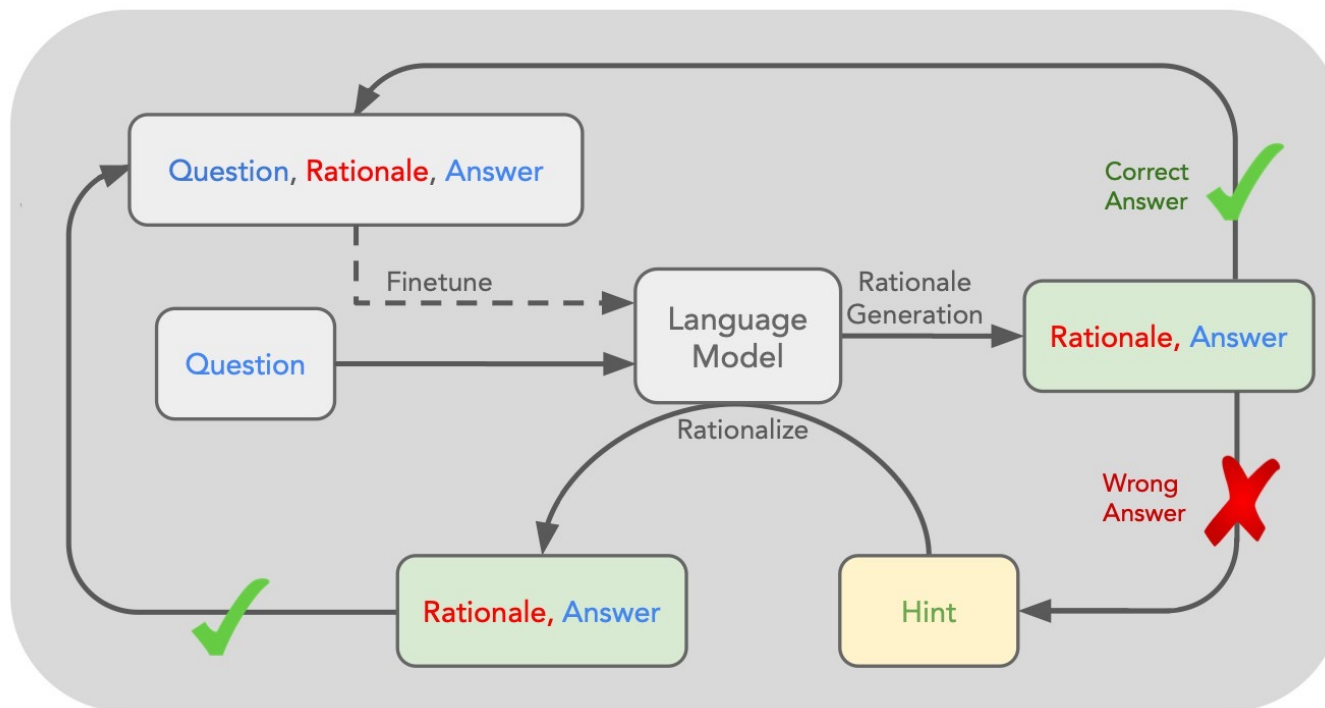


STaR: 自洽推理模型 Self-Taught Reasoner

基于推理过程的自举式推理优

[[Zelikman et al, 2022](#)]

Bootstrapping Reasoning With Reasoning



Q: What can be used to carry a small dog?
Answer Choices:
(a) swimming pool
(b) basket
(c) dog show
(d) backyard
(e) own home
A: The answer must be something that can be used to carry a small dog. Baskets are designed to hold things. Therefore, the answer is basket (b).

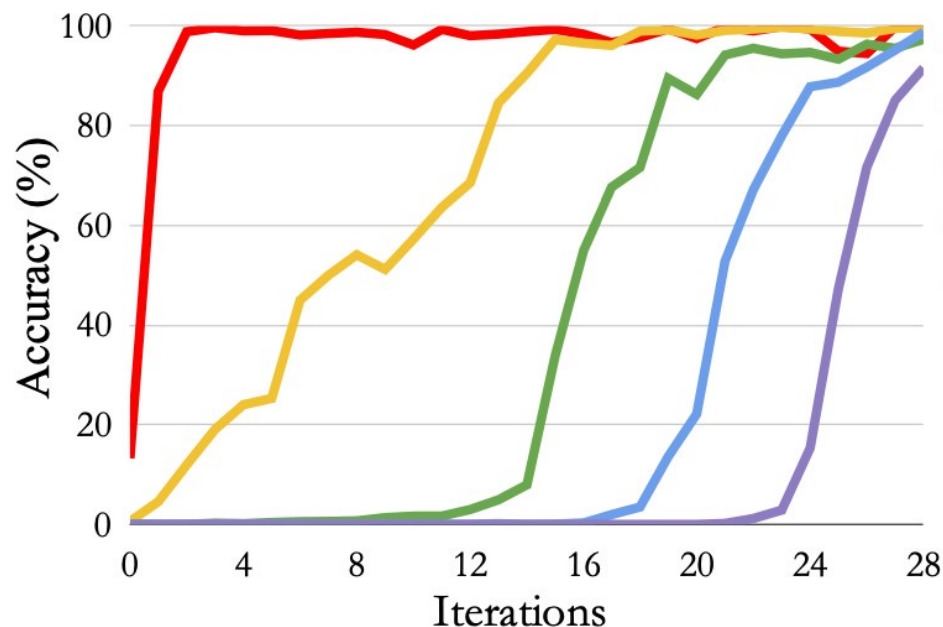
Figure 1: An overview of STaR and a STaR-generated rationale on CommonsenseQA. We indicate the fine-tuning outer loop with a dashed line. The **questions** and ground truth **answers** are expected to be present in the dataset, while the **rationales** are generated using STaR.

STaR: 自洽推理模型 Self-Taught Reasoner

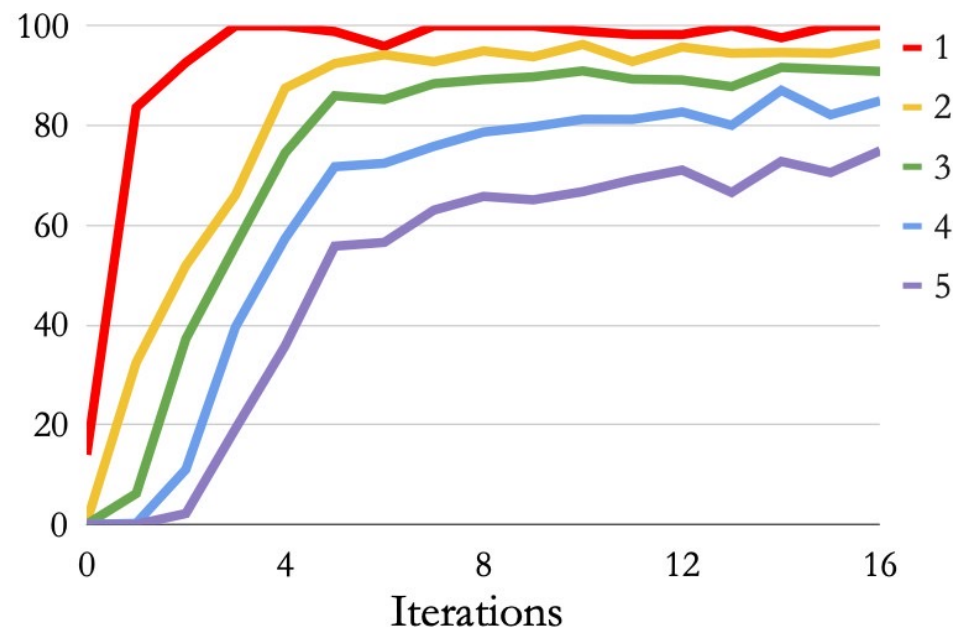
基于推理过程的自举式推理优

[[Zelikman et al, 2022](#)]

Bootstrapping Reasoning With Reasoning



(a) Without rationalization



(b) With rationalization

Figure 4: A visualization of the accuracy of n -digit summation with each iteration of STaR with and without rationalization for arithmetic. Each series corresponds to the accuracy of summing two n -digit numbers.

Quiet-STaR: 语言模型能够自主习得“先思后述”的能力

[[Zelikman et al. 2024](#)]

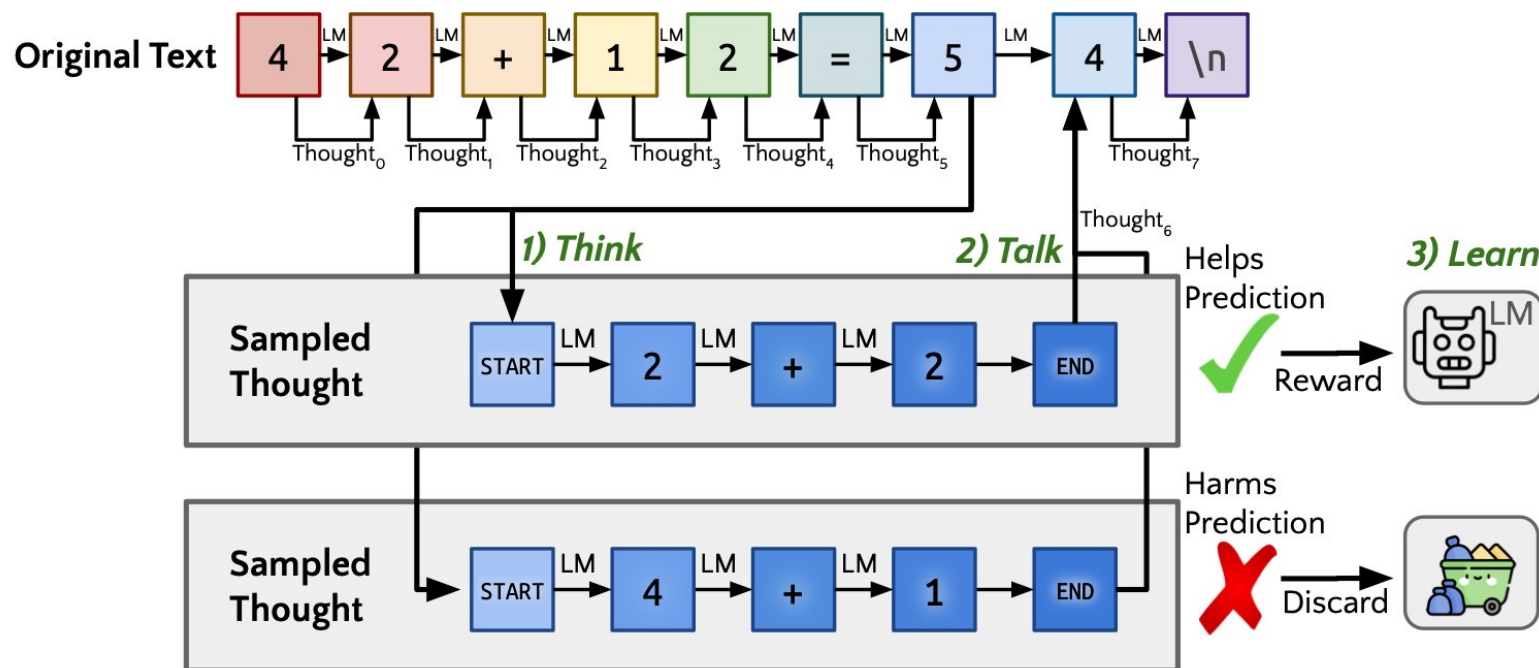
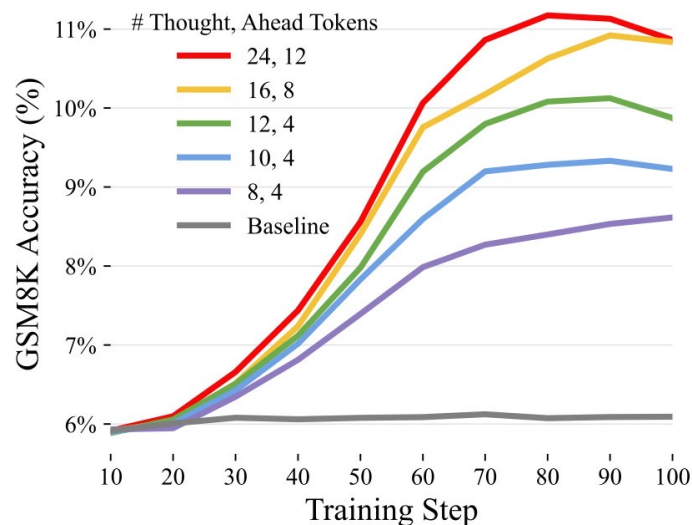


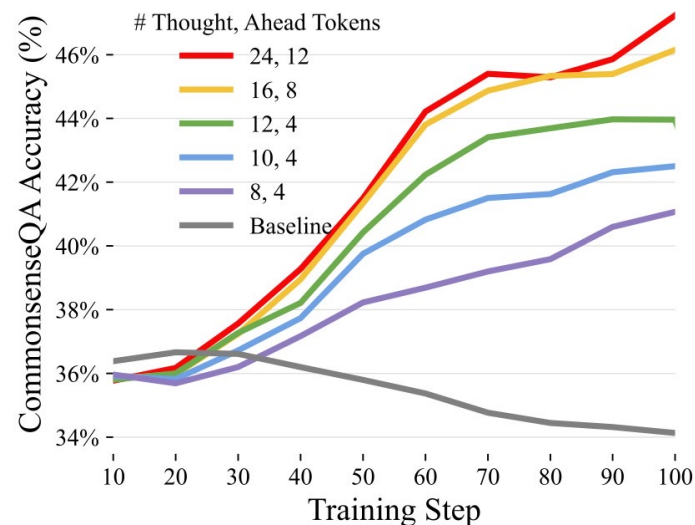
Figure 1: **Quiet-STaR**. We visualize the algorithm as applied during training to a single thought. We generate thoughts, in parallel, following all tokens in the text (**think**). The model produces a mixture of its next-token predictions with and without a thought (**talk**). We apply REINFORCE, as in STaR, to increase the likelihood of thoughts that help the model predict future text while discarding thoughts that make the future text less likely (**learn**).

Quiet-STaR: 语言模型能够自主习得“先思后述”的能力

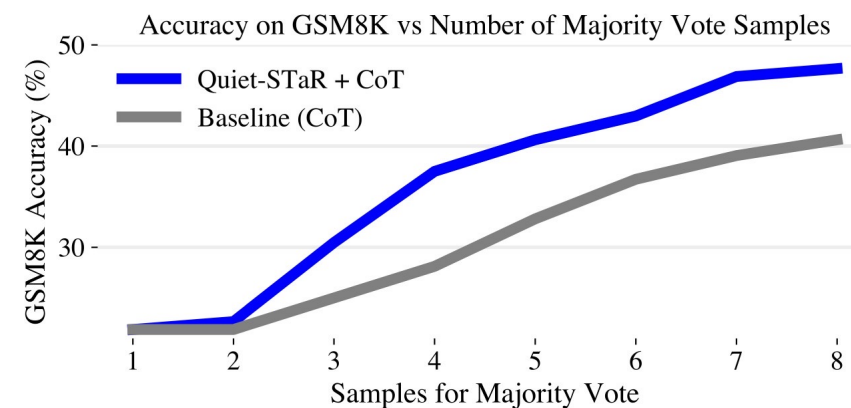
[[Zelikman et al. 2024](#)]



(a) GSM8K

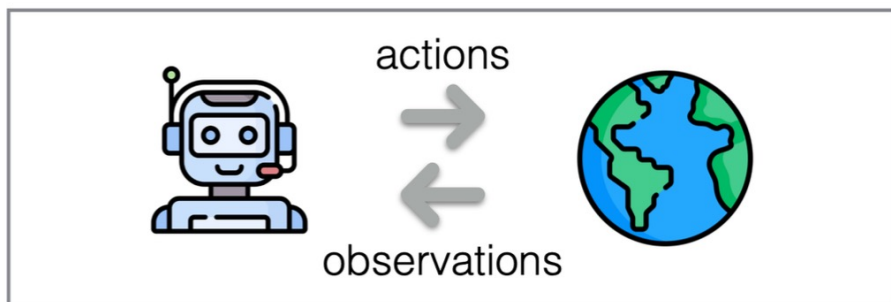


(b) CommonsenseQA



什么是大语言模型驱动的智能体？

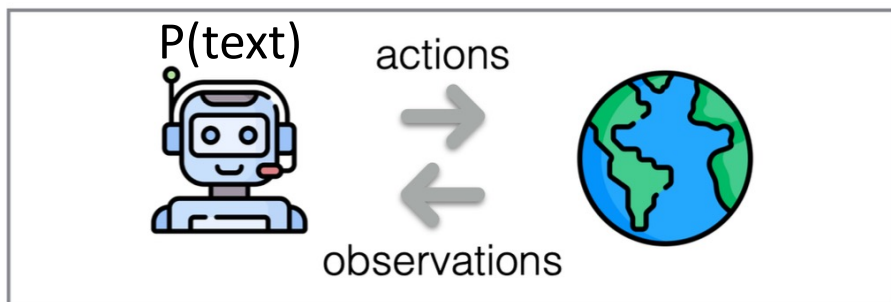
- 语言模型预测文本: $P(\text{text})$
- 智能体可在现实环境中迭代执行各类操作



- 大语言模型智能体的核心组件

- 基础大语言模型
- 提示词
- 动作/观测空间

- 语言模型智能体: 以语言模型为核心骨架的智能体



大语言模型的短板

- 数值/符号类操作

1. 计算
2. 逻辑推演
3. 精确操作

- 未纳入预训练语料的知识

1. 特定领域事实性知识
2. 全新信息
3. 私密信息

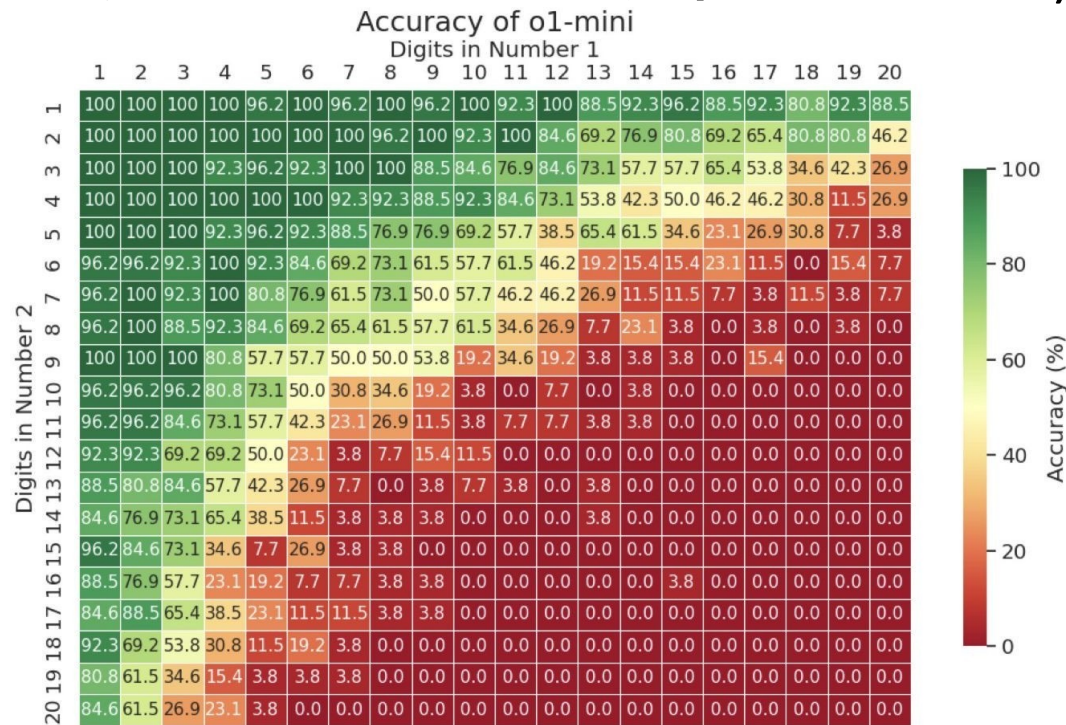
- 与外部环境的交互

1. 非自然语言接口
2. 物理世界
3. 环境信息(如时间)

针对大语言模型能力范围之外的任务，需为其提供辅助增强：
知识类、符号类及外部环境交互类任务

为什么用工具?

- 大语言模型并非万能的解决方案 (目前还远未达到AGI的水平，意外吗?)
- 即便是O1模型，也无法完成十位及以上数字的乘法运算……

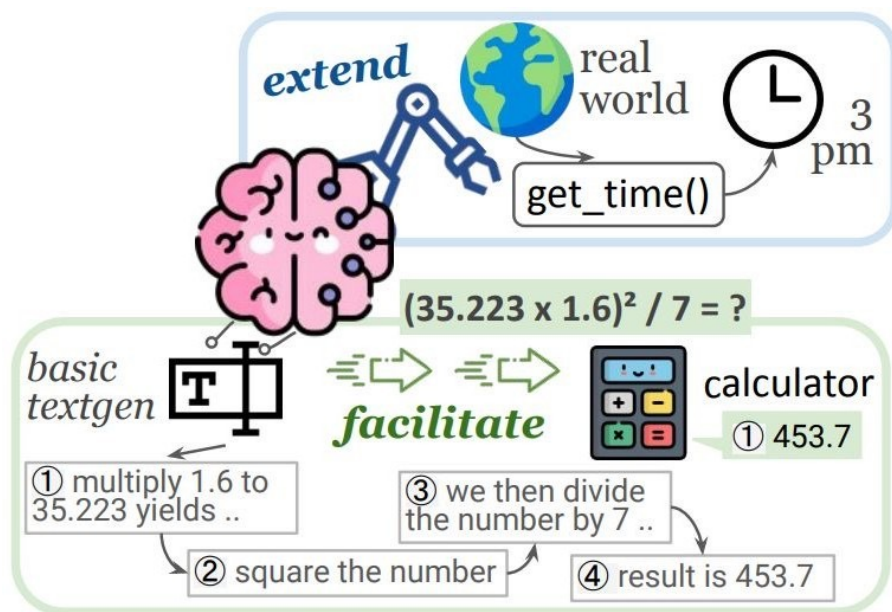


- 但我们凭什么指望大语言模型能做到这件事呢?
- 人类同样无法即时完成这类运算…… 但我们可以借助计算器轻松解决
- 那么，大语言模型也能使用工具吗?

什么是工具？

[[Wang et al. 2024](#)]

- **定义:** 语言模型调用工具，是指向一个独立于语言模型运行的计算机程序的函数接口—语言模型通过生成函数调用指令及输入参数，来实现对该工具的调用



工具的定义:

1. 一个计算机程序
2. 独立于语言模型运行
3. 通过生成函数调用指令实现调用

什么是工具?

[[Wang et al. 2024](#)]

Category	Example Tools
 Knowledge access	<code>sql_executor(query: str) -> answer: any</code> <code>search_engine(query: str) -> document: str</code> <code>retriever(query: str) -> document: str</code>
 Computation activities	<code>calculator(formula: str) -> value: int float</code> <code>python_interpreter(program: str) -> result: any</code> <code>worksheet.insert_row(row: list, index: int) -> None</code>
 Interaction w/ the world	<code>get_weather(city_name: str) -> weather: str</code> <code>get_location(ip: str) -> location: str</code> <code>calendar.fetch_events(date: str) -> events: list</code> <code>email.verify(address: str) -> result: bool</code>
 Non-textual modalities	<code>cat_image.delete(image_id: str) -> None</code> <code>spotify.play_music(name: str) -> None</code> <code>visual_qa(query: str, image: Image) -> answer: str</code>
 Special-skilled LMs	<code>QA(question: str) -> answer: str</code> <code>translation(text: str, language: str) -> text: str</code>

工具调用与智能体

- 智能体的定义

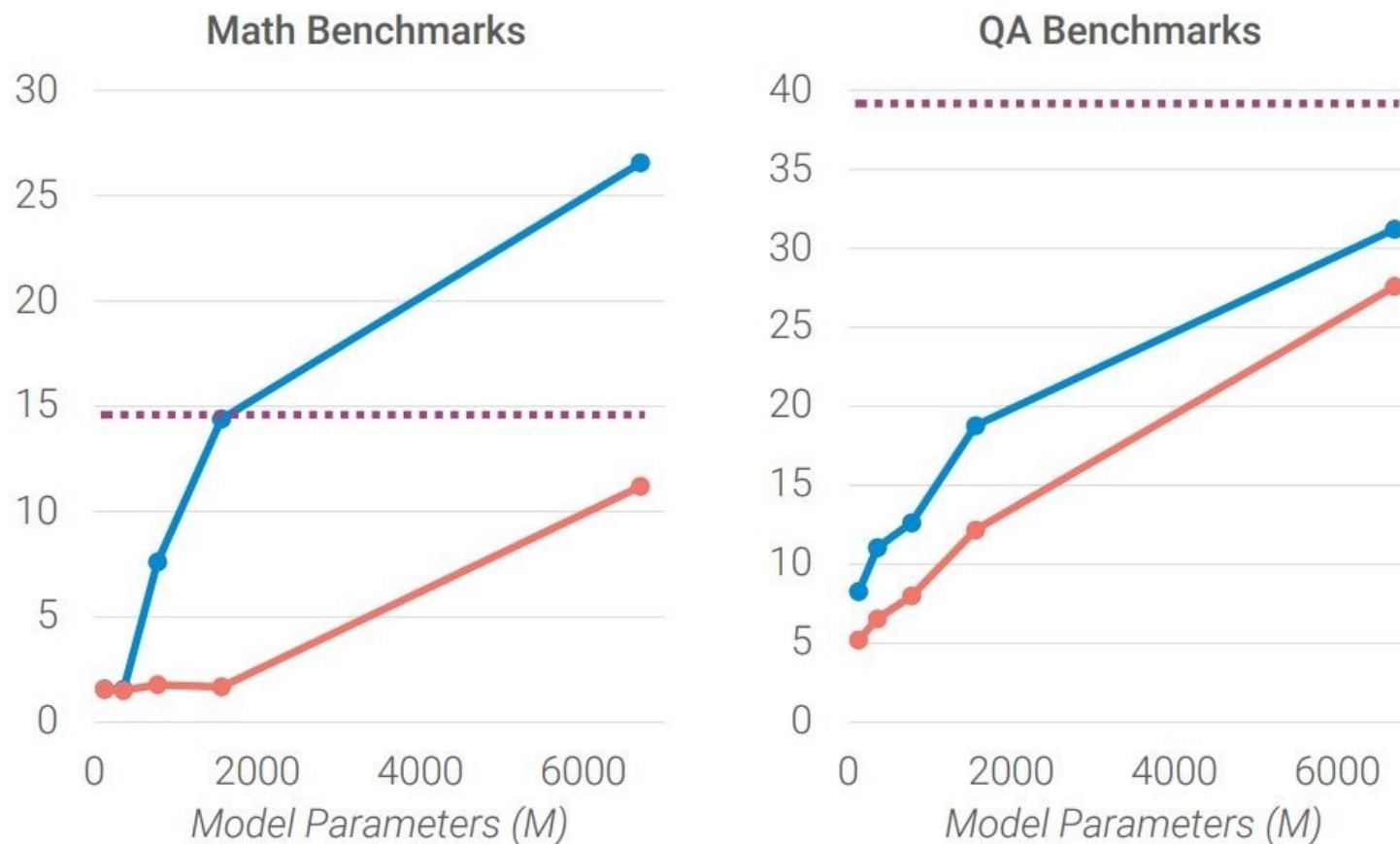
- 目前关于“agent”或“agentic”的定义尚未形成统一共识

- 核心要求:

1. *Probably*: 主动调用工具的能力
2. *Probably*: 具备迭代式的多步骤处理流程
3. *Maybe*: 具备与外部环境交互的能力

工具调用性能

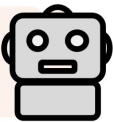
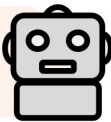
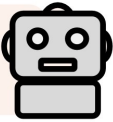
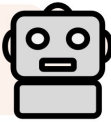
[[Snihck et al. 2023](#)]



显著提升GPT模型性能

模型能否主动提出澄清式问题？

- 这一点与人类相似，但语言模型在原始状态下，即便遇到指令模糊的情况，也不会主动提出疑问

 What is a good pasta recipe?	 What is a good pasta recipe?
 Cook pasta, add chicken broth... [wasted tokens]	 To start off, do you have any dietary restrictions?
 I am vegetarian! 🤔	 I am vegetarian. 🥑
 Here is a vegetarian... [relevant tokens]	 Here is a vegetarian... [relevant tokens]
Ineffective Conversation	Effective Conversation

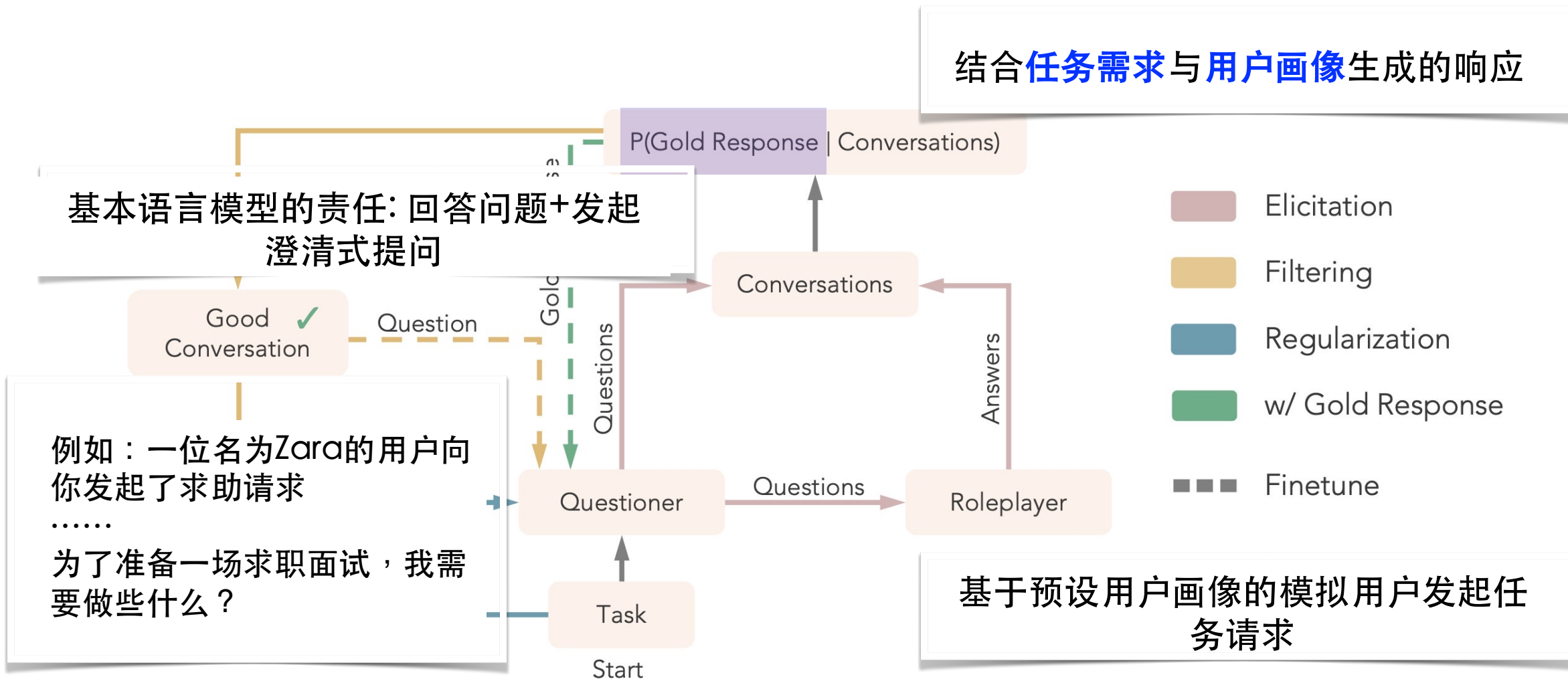
1. 任务歧义
2. 训练模型主动提出能够精准挖掘特定用户偏好的问题

[STaR-GATE, [Andukuri et al. 2024](#)]

STaR-GATE

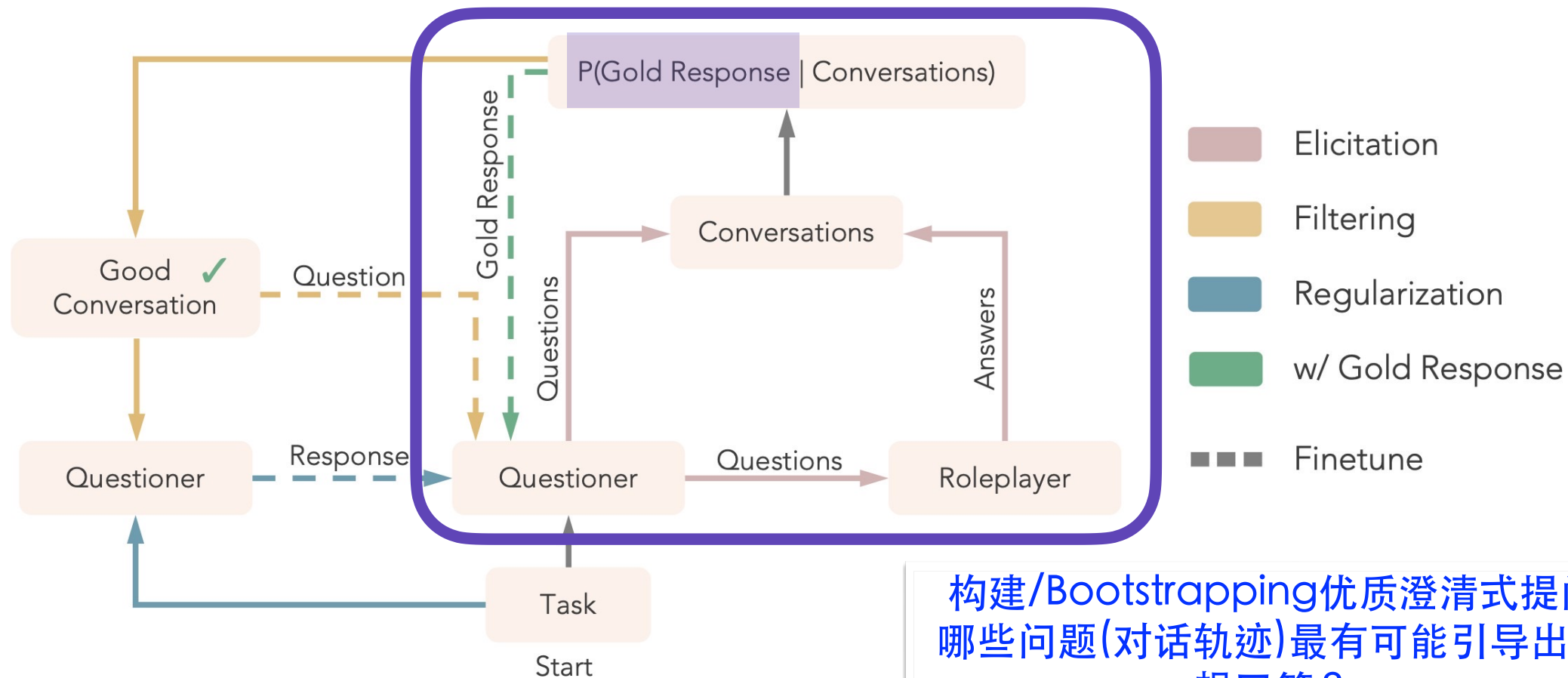
[STaR-GATE, [Andukuri et al. 2024](#)]

结合任务需求与用户画像生成的响应



STaR-GATE

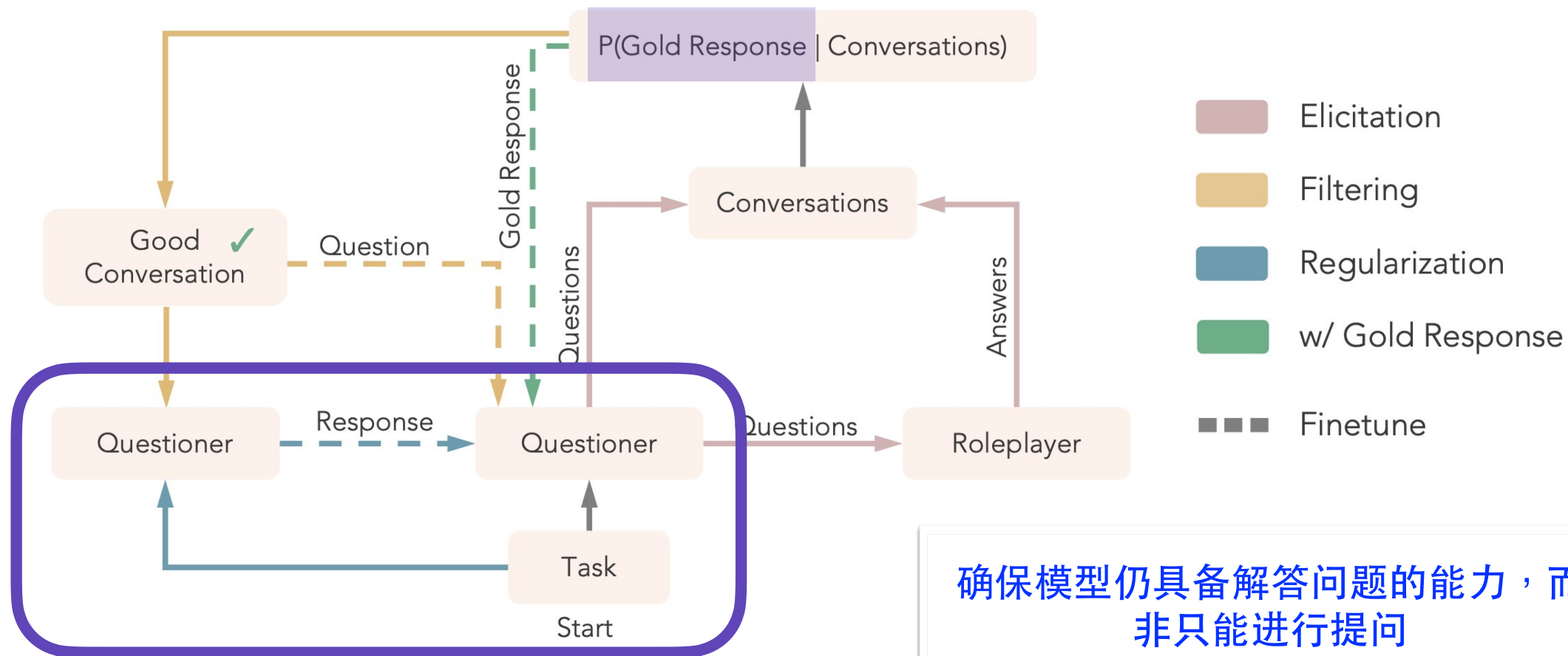
[STaR-GATE, [Andukuri et al. 2024](#)]



构建/Bootstrapping优质澄清式提问
哪些问题(对话轨迹)最有可能引导出理想回答？

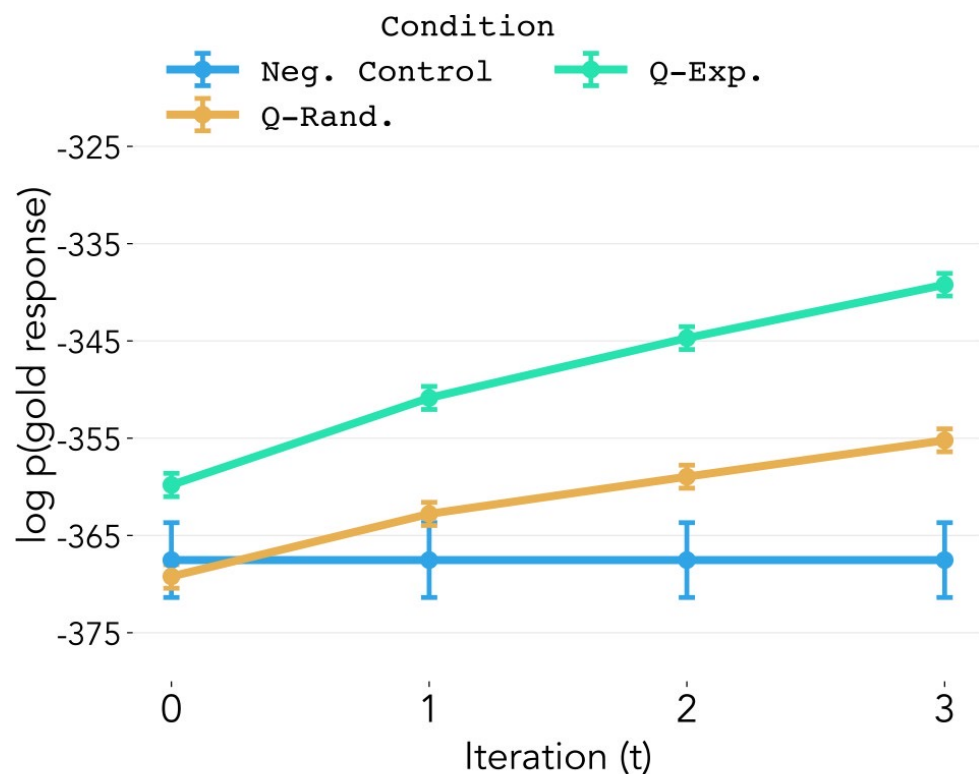
STaR-GATE

[STaR-GATE, [Andukuri et al. 2024](#)]

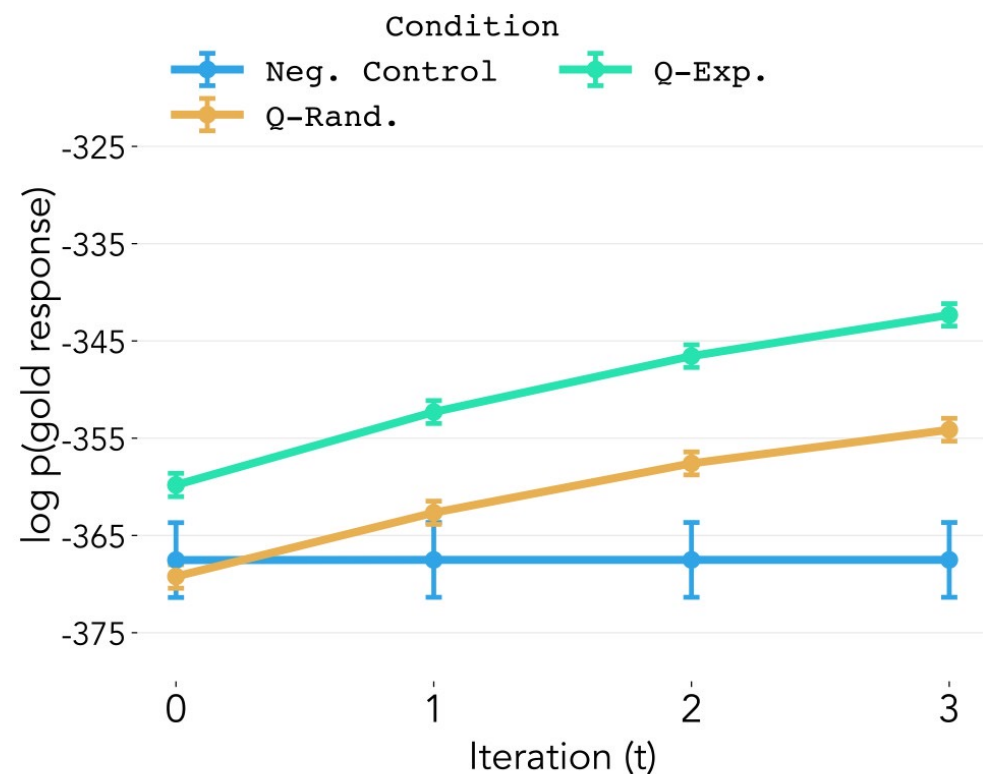


模型能否主动提出澄清式问题？

[STaR-GATE, [Andukuri et al. 2024](#)]

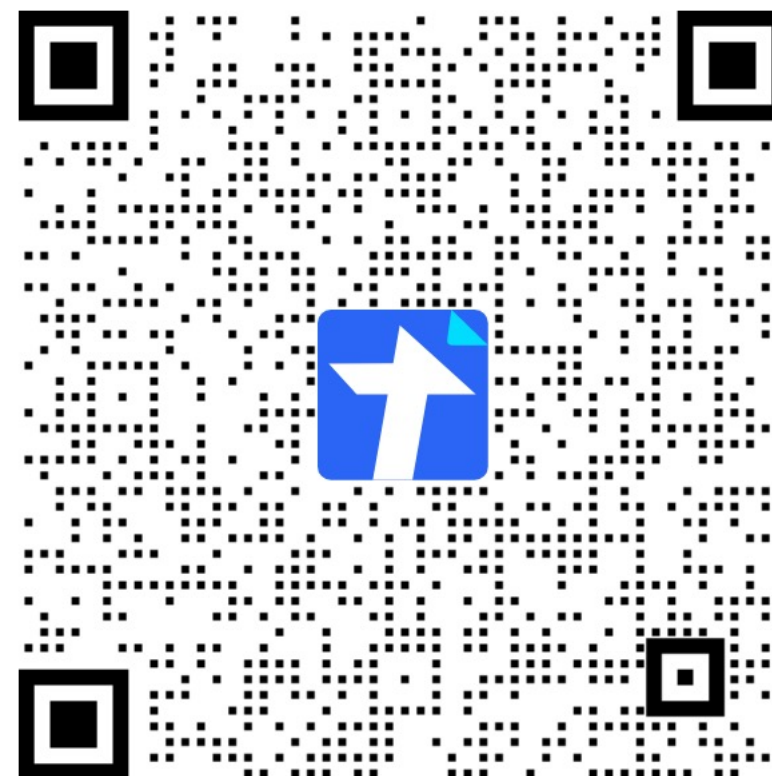


[a] STaR-GATE



[b] w/o Regularization

一句话总结



- 课外阅读
 - [SLP3] Ch. 7